

Validation of automated table extraction for historical infectious-disease surveillance data

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Abstract

The majority of historical epidemiological data remains undigitized. This paper evaluates the effectiveness of automated table extraction as an alternative to manual data entry. We assess the performance of *Amazon Textract* on a large dataset of historical infectious disease records, comparing extraction accuracy, error characteristics, and overall efficiency against traditional manual transcription methods. Cell-level agreement was evaluated on the 1956 tables (53,872 evaluated cells), where content-only agreement between the automated and manually transcribed data was 88.5% overall and 95.6% for numeric case counts; downstream epidemiological comparisons (disease peaks, growth rates, and cross-correlations) span all three years of the dataset, 1956–1958. Our findings show that extraction errors were present in all extractions we performed; they tend to be either minor with negligible impact or discernible on inspection. Despite requiring post-processing to address extraction errors, the automated approach substantially reduced estimated labour for these relatively regular printed tables, on printed, machine-typeset tables with a consistent weekly template. As well, while the post-processing time is slightly increased, it is a small proportion of the total

time previously required for the whole task. Overall, this paper demonstrates that automated extraction, when combined with targeted data processing, offers a viable solution for transforming historical tabular data into digital formats.

Author Summary

Vast amounts of historical public health data are stored in their original hardcopy form or in scanned images, so the data themselves are, in practice, inaccessible for statistical analysis. We set out to determine whether automated tools could help bridge this gap. We tested *Amazon Textract*, a machine learning service built specifically for structured document extraction, on historical Canadian infectious disease records. We found that after modest post-processing, the automated data closely matched records entered by hand: content-only agreement between the two sources (tolerant of formatting and of values shifted into an adjacent column) was 88.5% across all evaluated cells, rising to 95.6% for cells holding numeric case counts. Crucially, most errors that did occur differed from the reference value by only one or two characters and had negligible statistical impact; a small number of errors were large and occurred specifically in disease-name strings. We also showed that key epidemiological statistics, including disease peak timing, initial growth rates, and cross-correlations between diseases, showed only small, quantified differences whether calculated from the automated or manually entered data: peak lag was unchanged, peak correlation and peak week differed only slightly, and for the two diseases and epidemic windows we examined, the automated and manual growth-rate estimates were not distinguishable given the precision the data support (their 95% confidence intervals overlapped in every case tested). For researchers working with historical archives, this means that tools like Textract can meaningfully reduce the time and effort required to digitize data, freeing up resources for the scientific analysis itself, and facilitating larger data digitization projects. We hope this work encourages broader adoption of automated extraction methods in historical epidemiology and public health research.

1 Introduction

Weekly notifiable-disease reports have been collected in printed form for well over a century, but the overwhelming majority remain undigitized and therefore unavailable for computational analysis. Where such records have been converted to machine-readable form, this has typically been achieved by manual transcription – a process that is accurate but slow enough to constrain which archives are digitized at all, and which is not itself error-free (section 5 documents an instance in which the manual transcription used in this study contained an error that the automated extraction did not). Automated table extraction offers an obvious alternative, but for a researcher who intends to fit models to the resulting series, the relevant question is not how well a tool reads cells in the abstract. It is whether the epidemiological quantities computed from the extracted data match those computed from a trusted transcription, and, where they do not, whether the discrepancies can be detected without one.

Much work has been done on software that converts images of tables into machine-readable data, such as comma-separated value (CSV) files. The problems such software addresses are typically decomposed into distinct tasks, though different authors characterize them differently [1]. We organize our work around three tasks:

Table detection: Locating tables in documents.

Table structure recognition: Parsing the row and column layout of tables.

Content recognition: Identifying the content of table cells and headers.

Table detection and layout are surprisingly difficult tasks for computers given how easy they are for humans [2], a difficulty with roots in decades of OCR research [3] and still evident in widely used open-source engines such as Tesseract [4]. Modern document-AI systems – including large vision-language models applied directly to spreadsheets and tables – continue to report substantial failure modes on structurally irregular inputs even on contemporary,

cleanly scanned material [5, 6], a limitation that only compounds on historical documents, where period typography, physical degradation, and non-standard layouts are the norm rather than the exception. Software that actively produces usable results, in the sense of data similar enough to the scanned document to support similar downstream conclusions, must succeed at all three tasks at least reasonably well; a single under-performing sub-task is often enough to compromise the result, as we discuss in section 5.

Historical health-surveillance archives are a natural and comparatively under-served target for this technology. Project Tycho is the most developed prior effort of this kind, digitizing over a century of United States notifiable-disease reports [7, 8]; Canada’s own notifiable-disease surveillance system has a comparably long paper record that has, until recently, remained largely undigitized [9, 10]. What is largely absent from this literature, and from table-extraction benchmarks generally, is validation against an independent transcription with downstream, analysis-level epidemiological quantities – not only cell-level accuracy – as the endpoint. Benchmark accuracy on curated corpora does not say whether the peak timing, growth rate, or cross-correlation an epidemiologist computes from an extracted series would match what they would compute from a trusted transcription, or whether the discrepancies could be caught without one. Reference transcriptions are also conventionally treated as an infallible standard against which extraction is scored; in fact, as we show in section 5, they can themselves contain errors that a careful comparison can surface.

Using 156 weekly tables from the Canadian Disease Incidence Dataset (CANDID; 1956–1958) [11], we ask: how closely does a documented, staged Textract-based extraction and post-processing workflow reproduce an independently produced manual transcription at the cell level (evaluated here for the 1956 tables); which classes of error survive each processing stage; and how well are standard epidemiological summaries – peak timing, peak size, initial growth rate, and between-disease cross-correlation – preserved before and after targeted human review, for the disease pairs we examine? Useful table extraction tools must perform all three tasks above while balancing speed, accuracy, and cost against manual entry: extraction

time plus correction time should be smaller than the time to do the whole task manually, and the tool should be no more expensive. Even a tool that scored perfectly against these criteria would not obviate a researcher’s own reading of the results [12] – verifying a workflow like the one described here still requires a baseline of human time and attention.

Amazon Textract, the extraction service used throughout this study, is a cloud-based machine learning service that recognizes table, form, and key-value structure in scanned documents, in contrast to OCR tools that read only raw text; it is accessed either through a browser interface or programmatically via an API (section 2 gives implementation details).

2 Data extraction and Post-Processing

In this section, we will summarize the process of extracting data from a scan of a table to a CSV file, as well as the post-processing we found necessary to achieve usable results. Note that the details of the dataset we explored are not relevant to the basic extraction procedure. This process can be applied to any PDF repository of files (details on our specific dataset can be found in section 3.1). Information and specific details about Textract can be found in Amazon’s official documentation [13]. For supplementary material on running AWS Textract and the provided extraction script, see section 6.3.

2.1 API Call to Textract

AWS Textract can be accessed programmatically via API calls with any popular high-level programming language; we use Python, via AWS’s official SDK, `boto3`. A researcher calls Textract’s `AnalyzeDocument` endpoint and receives a flat, nested JSON structure of nested `BLOCK` objects (tables, cells, and words, each with a confidence score) that must be reassembled into the row/column structure a human would recognize; the full object structure is given in (S3). Once reassembled, this process can be applied to multiple tables at once by iterating through a repository of scanned table files and returning the outputs in separate

125 CSV files.

126 **2.2 Post-Processing**

127 Though it is trivial for humans to detect the difference between cell values and headers,
128 Textract relies on an OCR engine, so it does not differentiate between the two. Headers are
129 not explicitly labelled, merged cells rely on `RowSpan/ColumnSpan`, and column meaning must
130 be inferred by position. This is why post-processing is essential for meaningful analysis.

131 As our goal is to assess the quality of the extracted data, our post-processing involved
132 normalizing the cell data to match the original PDF tables' structure and cell types. This
133 involved trimming and removing whitespace, and converting cell values into strings.

134 There is also the issue of recognizing header values versus cell contents. This will, un-
135 fortunately, largely depend on the specifics of each dataset. For our purposes, we used the
136 metric of the proportion of cell values in a row that were numeric. This involves finding the
137 proportion of cell values that are strings in each row. Through manual inspection of the
138 original PDF dataset, we labelled rows with more than 50% of the cell contents being strings
139 as headers. In our tested datasets, this proved to be successful. In almost every extracted
140 table, the headers were removed successfully, with one exception that was manually caught
141 and edited.

142 Another major issue was rows being shifted by one cell value. This was problematic,
143 especially with numerical data, as it invalidates the results for an entire row when comparing
144 to the manual dataset cell by cell. However, this is also fairly easy to detect and fix when
145 we began comparing cells, so a step was added to check row length against the length of the
146 row above it. This was simple to program, and the results improved when we arranged the
147 data to be tested against the manual data. This is further described in section 4.

148 As well, in the data evaluation process, there were cells that were corrected to known
149 values based on the original dataset via dataset-specific coding. This process is described in
150 section 5.3.

3 Dataset Criteria and Selection

This section describes the dataset used to assess the quality of Textract-extracted data. Beyond the general tool-selection criteria set out in section 1 (accuracy, time, and cost relative to manual entry), the dataset itself had to satisfy one further, specific requirement: it had to have already been entered manually, so that the automated extraction could be checked directly against human data entry.

3.1 Canadian Disease Incidence Dataset

The Canadian Disease Incidence Dataset (CANDID) is a summary of cases of notifiable diseases in Canada, by province and territory, and was maintained by the Dominion Bureau of Statistics (the predecessor of Statistics Canada) [11]. The gathered data was stored in a given template and typed, so the majority of cell contents are easily legible to humans.

We chose this dataset for assessing Textract for a few reasons, but most importantly because it had already been independently transcribed by hand [11], which allows us to calculate the statistical measures of accuracy discussed in section 4. As detailed in section 4, we treat that manual transcription as a reference for comparison, not as a verified gold standard. We also chose this dataset for the qualities in the actual table, which we describe below.

We extracted data from the tables covering disease reports from 1956 to 1958 for a total of 3 years. There is a table for each week of the year, with columns corresponding to each province and territory recognized at the time, as well as totals for the country. Each section of columns corresponds to each province, which is further separated into the cases reported for the current week, the previous week, and the median value from the previous 5 years.

Each row of the table corresponds to a specific disease, for a total of 28 rows. Some rows correspond to only one disease, such as the first row corresponding to Chickenpox, while other diseases are split into multiple rows to store additional data. For example, Tuberculosis data

DOMINION BUREAU OF STATISTICS																																					
HEALTH AND WELFARE DIVISION																																					
Summary of Cases of Notifiable Diseases in Canada, as reported by the Provincial Health Departments																																					
for the week ended January 7, 1956, in comparison with the previous week ¹ , and the 3-year (1951-55) Median																																					
No.	DISEASE	CANADA ²					N.F.L.D.		P.E.I.		N.S.		N.B.		QUE.		ONT.		MAN.		SASK.		ALTA.		B.C.		No.										
		Report Week	Previous Week	Median 1951-55	Cumulative Total ³		Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week													
					1956	1955																															
1	Chickenpox	927	695	1,629	927	921	1,415	9	1	2	4	1	6	25	14	54	1	3	182	204	239	511	358	719	3	39	54	1	148	193	74	245	1				
2	Diphtheria	4	1	6	4	5	5	—	—	—	—	—	—	—	—	—	—	—	2	1	2	1	1	1	1	—	—	—	—	—	—	—	2				
3	Dysentery ⁴	11	4	9	11	4	13	—	—	—	—	—	—	—	—	—	—	—	3	1	3	2	1	1	1	—	—	—	—	—	—	6	1	2	3		
4	(a) Amoebic	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	4			
5	(b) Bacillary	11	4	9	11	4	12	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	6	1	2	2	
6	Encephalitis, infectious	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	6			
7	Influenza, epidemic	98	161	97	89	108	32	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	7			
8	Measles	1,974	999	1,955	1,974	1,975	1,375	55	11	11	69	99	1	86	44	106	2	10	95	63	197	108	242	453	11	36	41	1	5	89	47	6	99	22	30	272	8
9	Meningitis, meningococcal	4	9	4	4	2	4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	9		
10	Mumps	809	481	1,134	809	398	664	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	10		
11	Polioomyelitis, epidemic ⁴	1	6	5	1	5	5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	11		
12	(a) With paralysis	1	4	4	1	4	4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	12		
13	(b) Without paralysis	—	1	1	—	1	1	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	13		
14	Rubella	437	298	159	437	159	159	2	5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	14		
15	Scarlet fever	172	151	405	172	116	298	21	1	3	5	14	10	15	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	15		
16	Smallpox	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	16		
17	Tuberculosis ⁴	116	79	168	116	168	196	4	2	11	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	17		
18	(a) Pulmonary	105	67	133	105	134	141	4	2	10	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	18		
19	(b) Non-pulmonary	3	1	9	3	10	7	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	19		
20	Typhoid and paratyphoid	6	6	4	6	4	5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	20		
21	Undulant fever	3	1	1	3	1	1	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	21		
22	Veneral diseases:	259	240	416	259	396	336	6	2	11	1	8	94	37	117	37	46	58	10	33	38	12	10	33	39	36	51	22	23	24	25	26	27	28			
23	(a) Syphilis, primary	3	3	6	3	6	6	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	23		
24	(b) " secondary	3	—	1	2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	24		
25	(c) " other	20	20	43	20	38	39	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	25		
26	(d) Gonorrhoea	233	217	336	233	292	292	6	2	8	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	26		
27	(e) Other venereal diseases ⁵	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	27		
28	Whooping cough	151	118	190	151	331	159	2	5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	28		

(1) Figures adjusted for delayed reports. (2) Exclusive of Yukon and the Northwest Territories. (3) Total accumulated to the end of the current week from the first week ending in January of the year stated. (4) Includes cases where type was not specified. (5) Scarlet fever includes cases of streptococcal sore throat. (6) Type not specified. (7) Includes chancre, granuloma inguinale and lymphogranuloma venereum. ... Not reportable.

THE SITUATION IN THE UNITED STATES (Source: "MORBIDITY AND MORTALITY—WEEKLY REPORT", U.S. National Office of Vital Statistics: Number of reported cases for the current week (previous week in parentheses):
Diphtheria, 39 (54); Measles, 4,443 (3,725); Polioomyelitis, acute, 120 (119); Polioomyelitis, without paralysis, XX (XX); Smallpox, — (—); Undulant fever, 14 (14);
Encephalitis infectious, 20 (22); Meningococcal infections, 58 (77); Polioomyelitis, with paralysis, XX (XX); Scarlet fever, 31

Only 39 cases of diphtheria were reported this week as compared with a high of 104 for the week ended December 10, 1955. Almost half (17) of the cases were reported in the West South Central States. These and 2 Southern States, Alabama (8 cases) and Florida (4), accounted for all but 10 of the total cases.

EPIDEMIC REPORTS: Measles - Nfld. reports epidemic at Harbour Main.

9005-501
18-1-56

Figure 1: Week 1 table of the Canadian Disease Incidence Dataset. Source: Dominion Bureau of Statistics, 1956; reproduced under Crown copyright's 50-year term (expired at the end of 2006 for a 1956 publication), placing this material in the public domain in Canada.

on row 17 is where data is stored on Pulmonary and Non-Pulmonary Tuberculosis. See Week 1 of CANDID in fig. 1.

For a human observer, the table shown in fig. 1 is easily interpretable. However, from the perspective of an OCR engine such as Textract, some problems in the cell contents and structure arise. Below are the common problems seen in the cell structure and content that appear in many historical data documents (and lead to problematic results in section 5):

- Merged cells: In CANDID, many rows contain merged cells. For instance, the cells listing disease cases for Canada are merged, as are the province names and the header for the cumulative total. As a result, row lengths are inconsistent across the table.
- Header values that split the data for the current week, previous week, and median are printed vertically, whereas the rest of the table is printed horizontally.

- Where data was missing for the week, the document contains “..” as a placeholder.
- Where there are no cases, the document contains “-” as a placeholder.
- There is a lot of information useful for interpreting the data at the bottom of the table. Some cells have parentheses next to them with another number (ranging from 1 to 6) referring to a footnote. These footnotes correspond to how the data was collected, the province/territory it was sampled from, etc. For example, in fig. 1 the value “5(5)” is entered for P.E.I. under report week for the row “Scarlet Fever”. This corresponds to 5 cases of the disease in the province, and the (5) corresponds to the footnote “Scarlet Fever includes cases of streptococcal sore throat.”
- At the bottom of each week’s table there is information on data named, “The situation in the United States”. It is not stored in a tabular format, and was therefore not processed with Textract. This corresponds to Textract’s ability to successfully complete the table detection task described in section 1. However, this is valuable information that a statistician or researcher might need, so it is important to note that the tool we used did not process these data.

4 Statistical Measures

4.1 Testing Accuracy of Textract Data Compared to Manually Entered Data

To assess the accuracy of the CANDID data digitized by Textract, we compared it with the corresponding data digitized by humans in McMaster’s Theoretical Biology group, using a myriad of metrics.

We refer to this human-digitized dataset as the *reference transcription* rather than a verified gold standard. It is an independently produced transcription, not a ground truth:

like any manual entry process, it is itself subject to error, and we do not assume by definition that it is correct. Where possible, disagreements between the two sources are adjudicated against the original source scan rather than resolved in the reference transcription’s favour (see section 5.3 for an example in which this adjudication instead identified an error in the reference transcription).

4.1.1 Simple Metrics

The most straightforward metric, which we call the *exact matched-cell proportion*, is the percentage of cells from the Textract dataset that are exactly the same as the reference transcription. This is not a measure of cells that were correctly digitized, but rather the proportion of cells whose both content and location (row and column index) are equivalent.

This percent equivalent metric was first applied to the raw extracted data, and then again as the data became progressively more processed through the methods described in section 2.2.

Inclusion rule. We define the evaluated cell set reproducibly, following the implementation in `postprocessing/combine.ipynb`: each cleaned weekly CSV pair (`data/textract_cleaned/1956`, `data/manual_cleaned/1956`) is compared position-by-position over the overlapping row and column range, i.e. `min(n_rows)` by `min(n_cols)` between the two files for that week; any rows or columns beyond that overlap (e.g. from a length mismatch between the two sources) fall outside this range and are not included in the denominator. Merged-cell handling, header/data-row labelling, footnote-marker treatment, and repeated row indices are resolved upstream during the cleaning stage (section 2.2) that produces the `*_cleaned` inputs, not by the comparison script itself; the comparison operates on whatever string is present in each cleaned cell. Blank cells and “-”/“..”/“...” placeholders are retained and compared as literal strings. A cell counts as numeric (and is included in the numeric-only denominator in table 1) if both the Textract and reference values consist only of digits with optional

thousands separators; cells where either side is empty or a placeholder are excluded from the numeric-only comparison. The row-shift tolerance (table 1, Stage 2) checks whether a Textract cell matches the reference either in its original column or in the adjacent column one position to the right.

4.1.2 String Differences and Levenshtein Distance

To quantify discrepancies between cells with extracted string and their manually entered counterparts, we compute string-level similarity metrics, with a primary focus on Levenshtein distance [14]. Levenshtein distance measures the minimum number of single-character edit operations – insertions, deletions, or substitutions – required to transform one string into another; a distance of zero indicates an exact match, while larger values reflect increasing divergence. The standard dynamic-programming recurrence for computing it is given in (S4).

The dynamic-programming formulation above follows the general string-to-string correction framework of Wagner and Fischer [15]. In practice, Levenshtein distance is often normalized by string length to enable fair comparison across tokens of different sizes. This is particularly important when evaluating OCR output, where short strings (e.g., years or counts) are disproportionately sensitive to single-character errors.

4.2 Epidemiologically Significant Statistics Derived from Textract Data

4.2.1 Row and Column Sums

Aggregate counts obtained by summing across diseases or locations are often needed. If these sums were provided in the original documents, then they also provide a valuable opportunity for consistency checks in the data entry. Thus, it is extremely useful to sum values across a row or down a column and confirm that the result matches the reported total at the

end of that row or column. If the sums agree, it provides an immediate check that entries were not missed, duplicated, or mistyped. While it is technically possible for errors in both the extracted sums and incidence values to cancel out and still agree, this is very unlikely. Discrepancies immediately flag rows or columns that require closer review, allowing errors to be identified and corrected early before downstream analysis.

4.2.2 Cross-Correlation Between Diseases

Computing cross-correlation between two disease time series involves measuring how strongly their observed patterns move together over time, and whether changes in one disease tend to occur before, after, or at the same time as changes in the other. Instead of determining whether two diseases are correlated in a broad sense, cross-correlation examines their relationship across different time shifts. For example, whether peaks in Measles are followed by peaks in Chickenpox one or two weeks later. Mathematically, it compares the standardized values of one time series with shifted versions of the other, producing a correlation value for each lag.

Ordinary correlation measures the linear association between two random variables X and Y . It is defined as the normalized covariance. This quantity lies in $[-1, 1]$ and measures the strength and direction of linear dependence.

For time series $\{x_t\}$ and $\{y_t\}$, ordinary correlation corresponds to comparing values at the same time index:

$$\rho_{xy}(0) = \frac{\text{Cov}(x_t, y_t)}{\sigma_x \sigma_y}.$$

Cross-correlation extends this idea by introducing a time shift, which is called the lag. The cross-covariance at lag k is

$$\gamma_{xy}(k) = \text{Cov}(x_t, y_{t-k}),$$

280 and the cross-correlation function (CCF) is

$$\rho_{xy}(k) = \frac{\gamma_{xy}(k)}{\sigma_x \sigma_y}.$$

281 Thus, $\rho_{xy}(k)$ is the correlation between x_t and a shifted version of y_t . By varying k ,
 282 cross-correlation measures linear dependence across different time offsets.

283 **Interpretation:**

- 284 • $k = 0$: simultaneous association.
- 285 • $k > 0$: x leads y by k time units.
- 286 • $k < 0$: y leads x by $|k|$ time units.

287 We will compute the cross correlation function, CCF, $\rho_{xy}(k)$ with x and y , referring to
 288 two different diseases that were both reported during our time period. Our objective is
 289 to compute the CCF using the manually curated data, and then compute the same CCF
 290 using the automatically extracted data. This produces two cross-correlation sequences corre-
 291 sponding to the manual and extracted datasets, respectively. If the two CCFs are sufficiently
 292 similar, we will conclude that the automatic data extraction is sufficiently accurate to make
 293 epidemiologically meaningful inferences.

294 To assess similarity, we compare these functions across all lags k . This can be done
 295 by examining differences in (i) the lag at which the maximum correlation occurs, (ii) the
 296 magnitude of peak correlations, and (iii) the overall shape of the CCF curves. Quantitatively,
 297 one may compute the difference

$$\Delta(k) = \rho_{xy}^M(k) - \rho_{xy}^T(k),$$

298 where superscripts M and T refer to manual and Textract data, respectively. If the two
 299 CCFs are similar in magnitude, peak location, and overall structure, this suggests that the

extracted data preserves the dependence structure present in the manual dataset.

4.2.3 Other Time Series Statistics

Another verification tactic is to filter the data by disease—or more generally, by category or type—and examine each subset independently. Analyzing grouped data makes it easier to spot values that are inconsistent with the typical range for that specific group. Outliers can then be identified by visual inspection of scatterplots, boxplots, interquartile range, or standard deviation-based cutoffs.

Using more than one approach reduces the risk of missing meaningful anomalies or over-flagging valid but extreme observations, and helps distinguish true data entry errors from legitimate but rare values.

The test statistics of greatest interest are those that an epidemiologist would be naturally interested in calculating. In a time series of disease incidence, the *peak height* is simply the largest observed value (the maximum number of cases), and the *peak time* is the time index or date at which this maximum occurs. For a time series of a given disease, knowing the peak height and the time at which it occurs is important as it helps predict the intensity of an outbreak.

Peak-detection rule (tables 4 to 5). Because a raw weekly maximum is sensitive to isolated OCR spikes and single-week noise, peaks are located from a locally smoothed series rather than the raw weekly values: within each fitting window, we fit a local quadratic regression (LOESS, span 0.4, degree 2) to weekly cases against week index, and report the week and fitted value at which this smoothed curve is maximized. Ties in the smoothed maximum resolve to the earliest week in the window. Weeks with a missing time index or missing case count are dropped prior to smoothing rather than imputed; a window is not fit if fewer than 5 non-missing weeks remain. We separately checked whether any reported peak week (Measles: weeks 64 and 126; Chickenpox: weeks 64, 112, and 114) coincides with

a week flagged by the row-sum validation of section 5.3: none can be checked, because that validation covers only the 52 weeks of 1956, while every reported peak here falls in 1957 or 1958. This is itself a limitation of the current validation coverage, not evidence that the reported peaks are error-free.

In the context of analyzing the quality of our extracted data, testing the growth rate is important as it is one of the preliminary statistics that researchers want to know about an epidemic. Estimating the initial growth rate of a disease time series quantifies how quickly the number of cases increases during the early phase of an outbreak. The growth rate provides a concise measure of epidemic speed and can be used to compare outbreaks across locations or time periods. The early phase of an outbreak is typically well-approximated by exponential growth, where the expected number of cases increases proportionally to the current number of cases. The initial exponential growth rate λ is particularly important as it yields the epidemic doubling time ($\frac{\ln 2}{\lambda}$); in principle it can also be used to derive quantities such as the basic reproduction number $\mathcal{R}_0$ [16], though we do not compute $\mathcal{R}_0$ in this paper and mention it only as context for why r is a quantity of independent epidemiological interest, not only an intermediate step toward the doubling time.

Methods for estimating the initial exponential growth rate [17, 18] typically take advantage of the fact that if incidence is growing exponentially then cumulative incidence also grows exponentially. Since datasets are often observed at substantial intervals (e.g., weekly), designing methods that use “interval incidence” (the difference between cumulative incidence at observation times) is more robust. We can also use the growth rate to calculate the doubling time, which is a statistic that is more easily interpreted numerically. The exponential growth rate r is related to the doubling time T_d by

$$T_d = \frac{\ln 2}{r}$$

Model specification (tables 4 to 5, figs. 15 to 16). Growth rates and doubling times for Measles and Chickenpox were estimated with `epigrowthfit` 0.15.5 [18] in R 4.5.1, using `egf_model(curve = "logistic")` with all other model arguments left at their package defaults (negative-binomial dispersion, no day-of-week effect – not applicable, since our data are weekly rather than daily – and no excess-cases term). Every top-level model parameter, including $\log(r)$, used the default intercept-only formula, so a single growth rate r was estimated jointly from both fitting windows for a given disease and data source, rather than a separate rate per window; section 5 reports what this implies for the two-window presentation in figs. 15 to 16. Confidence intervals are Wald intervals computed on the $\log(r)$ scale via `confint.egf()` and back-transformed. The two fitting windows (week 35–64 and week 95–126, bracketing the two annual epidemic waves visible in the incidence curves) were chosen by visual inspection of the manually entered time series and then applied unchanged to the corresponding Textract series for the same disease; windows were not re-chosen from the Textract data itself. Weeks with a missing time index or case count inside a window are dropped prior to fitting, matching the treatment used for peak detection above.

5 Analyzing the results from the Canadian Disease Incidence Dataset

The following section explores the accuracy of table extraction on a particular dataset, namely the Canadian Disease Incidence Dataset (CANDID).

5.1 Comparison to Manually Entered Data

5.1.1 Simple Metrics

The cell-agreement analysis in this section covers the 1956 CANDID tables only, not the full 1956–1958 (156-table) dataset described in section 3.1. It was computed with the eval-

371 uation script in `postprocessing/combine.ipynb`, run over the 52 weekly tables for 1956
 372 (53,872 evaluated cells in total; 26,069 of these hold numeric values on both the Textract
 373 and reference sides). When comparing values cell by cell, the percentage of cells between
 374 the raw Textract data and the manually entered reference transcription that were equivalent
 375 (both content and position, i.e. strict string equality; see section 4) was 74.0% across all
 376 evaluated cells, or 77.9% when restricted to the numeric-only cells. Applying a sequence
 377 of processing stages to the Textract output increased content-only agreement (tolerant of
 378 comma/whitespace formatting and of a value being shifted into an adjacent column) to
 379 88.5% across all cells, or 95.6% when restricted to numeric-only cells. Table 1 summarizes
 380 each stage, its resulting agreement under both denominators, and, critically, whether the
 381 underlying rule was fixed independently of this dataset (*a priori*), developed by inspecting
 382 the reference transcription, or applied as a manual, post-hoc correction. This distinction
 383 matters because rules developed by inspecting the reference data are not validated against
 384 unseen material (see the leakage discussion in section 6.2).

Stage	Operation	All cells (n=53,872)	Numeric cells only (n=26,069)
0	Raw Textract output (strict equality)	74.0%	77.9%
1	+ Comma/whitespace normalization	76.8%	83.8%
2	+ Row-shift tolerance (adjacent column)	88.5%	95.6%
3	Manual adjudication (Meningitis, Mumps)	not quantified	not quantified

Table 1: Pipeline stages applied to the raw Textract output for the 1956 CANDID tables, and the resulting cell-agreement percentage at each stage under two denominators (all evaluated cells vs. numeric-only cells), as computed by `postprocessing/combine.ipynb`. Stage 2 reports content-only agreement (a match either in place or shifted one column over); it is not a strict content-and-position metric. The numeric-only column is not a denominator change applied to Stage 1’s value but an independently recomputed percentage over the numeric-only subset.

385 5.1.2 String Differences

386 Using Python’s NLTK library and its Levenshtein distance function, it is relatively straight-
 387 forward to find the Levenshtein distance between two strings. The goal is to find the distri-

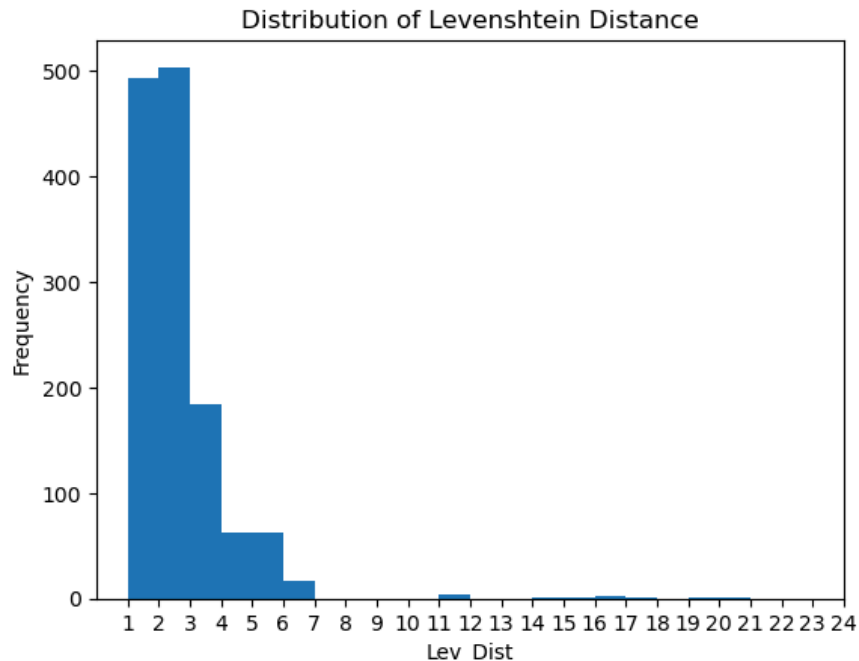


Figure 2: The frequencies of Levenshtein Distances (defined in section 4.1.2) between manual and Textract cell values in the CANDID. The majority of distances measured are 1 or 2 differences in characters between the two strings.

bution of edit distances among mismatched entries, from cells differing by a single character to cells with little resemblance to the reference value. It is illustrative to find the average distance from entries where an actual error occurs. Recall from section 5.1.1 that content-only agreement (tolerant of comma/whitespace formatting and of a value being shifted into an adjacent column) is 88.5% across all evaluated cells; this is distinct from the strict content-and-position agreement of 74.0%, in which a cell’s content may be correct but extracted to the wrong location within a table. In the analysis below, “mismatched” cells are those failing even the content-only criterion (`Any_Equality = False` in the evaluation script), i.e. the residual disagreements not resolved by any of the processing stages in table 1. The average Levenshtein distance among these mismatched cells is approximately 1.84. Separately, the modal distance is 2 (51.7% of mismatched cells), with 1 the next most common (38.1%). fig. 2 shows the proportion of cell contents per Levenshtein distance.

We see the most common Levenshtein distances are as expected, namely distances of 1

401 or 2. This aligns with results from section 5.1.1, where most cell contents differ by a single
 402 digit or special character. Of the tables for 1956, 6,177 of 53,872 evaluated cells (11.5%) were
 403 cells that we considered to have true extraction errors (i.e. mistakes that were not resolved
 404 through post-processing); of these, 38.1% have a Levenshtein distance of 1 and 51.7% have
 405 a Levenshtein distance of 2.

406 The distance value then drops off steeply after a distance of 2, with a thin tail extending
 407 out to a maximum of 25. It is likely the OCR engine confuses a blurry part of a number, or
 408 that the structure of the table or quality of scan precludes accurate extraction of one or two
 409 digits.

410 The percentage of each Levenshtein distance in cells with errors from the tables for 1956
 411 is seen in table 2.

Levenshtein Distance	Count	Percentage
1	2356	38.14%
2	3194	51.71%
3	314	5.08%
4	98	1.59%
5	114	1.85%
6	84	1.36%
8	3	0.05%
9	2	0.03%
11	7	0.11%
14	4	0.06%
25	1	0.02%

Table 2: Distribution of Levenshtein distances between extracted and reference values, computed by `postprocessing/combine.ipynb` over the 1956 CANDID tables. Counts sum to 6,177 cells with true extraction errors, out of 53,872 evaluated cells.

412 There are outliers in the trend seen at Levenshtein distances of 8, 9, 11, 14, and 25. These
 413 errors are almost certainly specific to this dataset and processes. For example, the single
 414 distance-25 outlier corresponds to a cell where Textract returned a run of placeholder dots
 415 (.....) instead of a value, compared against the reference value
 416 68: the entire reference string had to be inserted character-by-character, producing a large
 417 Levenshtein distance. The specific Levenshtein distances, and where they are located, would

depend on the structure and content of the table in question. We can expect that longer strings, or cells where Textract fails to extract any numeric content at all, would contribute to a larger Levenshtein distance.

For Levenshtein distances greater than 6, the errors we inspected occur in the disease-name column (column index 0) of the affected tables, where Textract extracted a word or word-fragment (e.g. “meningococcal”, “infectious”, “epidemic”, “primary”) into a cell the reference transcription records as a plain numeric value or placeholder. This pattern is consistent with row misalignment: a multi-word or multi-row disease-name label bleeding into the row Textract assigned to the adjacent numeric entry.

Of course, Levenshtein distances of large magnitudes help us in the sense that, following a manual inspection of the extracted data, it would be clear to a human that an error has occurred. In addition, distances of this magnitude occurred exclusively in disease-name strings rather than in numeric values.

We also looked at the relative Levenshtein distance, i.e., the Levenshtein distance proportional to the length of the reference string. This allows us to visualize the errors caused by Textract in a way that accounts for the length of the string (see fig. 3). As longer strings have a higher potential Levenshtein distance, normalizing by their lengths aids in determining the proportion of mistakes.

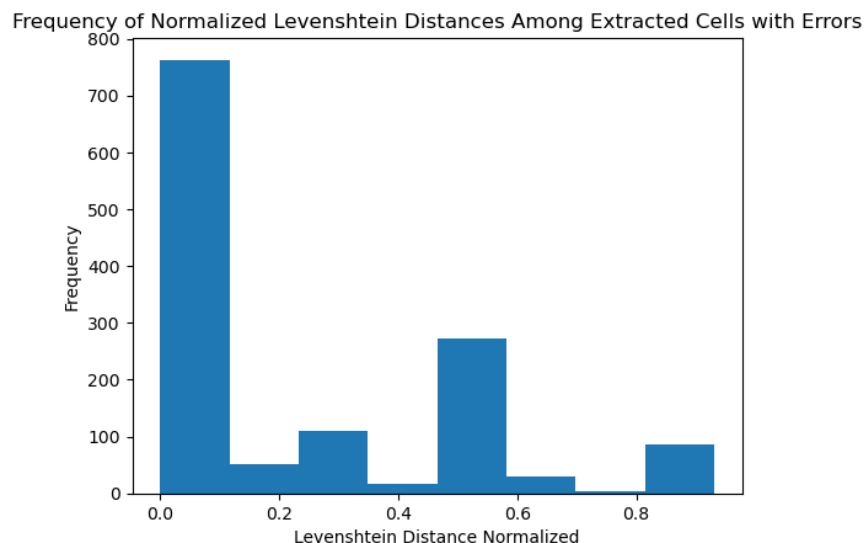


Figure 3: Levenshtein Distance (as defined in section 4.1.2) normalized by the length of the string of the manually entered cell value.

In cells with errors, we see that the most common values of normalized Levenshtein distance is 0.1. The other spikes in the graph, around 0.5 and greater than 0.9 can be attributed to shifted cells. For example, when Textract inputs the name of a disease in a cell location that is expected to be a disease count, the character counts between the Textract and manual cell values will vary greatly.

5.1.3 Detecting Minor Errors with Row Sums

To mitigate errors arising from row misalignment during extraction, rows corresponding to the same disease were first grouped from each table. Because similar shifting occurs for the same disease in each table, grouping rows by disease prior to checking row sums makes it possible to enforce consistent indexing and ensure that computed quantities are evaluated over the correct set of entries. The structure of the data used for this computation is illustrated in fig. 4. The total incidence value for the week is in a column titled “report week” under “Canada” and the contributing cell values similarly appear every third column for each province and territory.

DOMINION BUREAU OF STATISTICS																							
HEALTH AND WELFARE DIVISION																							
Summary of Cases of Notifiable Diseases in Canada, as reported by the Provincial Health Departments																							
for the week ended March 3, 1966, in comparison with the previous week ¹ , and the 5-year (1961-5) Median																							
No.	DISEASE	CANADA ²				N.F.B.		P.E.I.		N.S.		N.B.		O.N.T.		M.A.N.		S.A.S.		A.L.T.A.		B.C.	
		Report Week	Previous Week	Median 1961-5	Cumulative Total ³	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week	Report Week	Previous Week
		1966	1965	1961-5	1966	1966	1965	1966	1965	1966	1965	1966	1965	1966	1965	1966	1965	1966	1965	1966	1965	1966	1965
1	Chickenpox	1,874	1,127	1,127	11,139	10,139	13,139	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024	
2	Diphtheria	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	
3	Diphtheria *	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	
4	(a) Ankle	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	
5	(b) Ankle	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	
6	Herpes zoster	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
7	Influenza, epidemic	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
8	Measles	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
9	Meningitis, meningococcal	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
10	Mumps	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
11	Poliomyelitis, epidemic *	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
12	(a) With paralysis	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
13	(b) Without paralysis	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
14	Rubella	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
15	Scarlet fever, etc. *	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
16	Smallpox	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
17	Tuberculosis *	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
18	(a) Pulmonary	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
19	(b) Non-pulmonary	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
20	Typhoid and paratyphoid	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
21	Typhoid fever	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
22	Veneral diseases:	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
23	(a) Syphilis, primary	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
24	(b) " secondary	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
25	(c) " other	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
26	(d) Gonorrhea	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
27	(e) Other venereal diseases *	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	
28	Whooping cough	1,072	1,072	1,072	10,720	10,720	10,720	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	1,072	

(1) Figures adjusted for delayed reports. (2) Exclusive of Yukon and Northwest Territories. (3) Total accumulated to the end of the current week from the first week of the year. (4) Includes cases where type not specified. (5) Scarlet fever includes cases of streptococcal sore throat; streptococcal sore throat not reportable in New Brunswick and Quebec. (6) Type not specified. (7) Includes gonorrhea, chlamydia, trichomonas, and other venereal diseases. (8) Includes cases where type not specified.

Figure 4: Rows contributing to the computed sums are in blue, and the expected sums are in pink, of the tables in CANDID section 3.1. In the original PDF table, we expect that across a row, the sum of each value in the blue columns equals the total in pink.

By disease, we plot the difference between the expected sums, as extracted by Textract, and the actual sum calculated by looping over the row contents. Below are the values calculated for Chickenpox (see fig. 5).

Most rows exhibit zero difference, indicating agreement between extracted totals and re-computed values. Nonzero deviations indicate inconsistencies that can be traced to numerical errors in individual cells.

For example, the first outlier in fig. 5 corresponds to Week 9 in the dataset. From

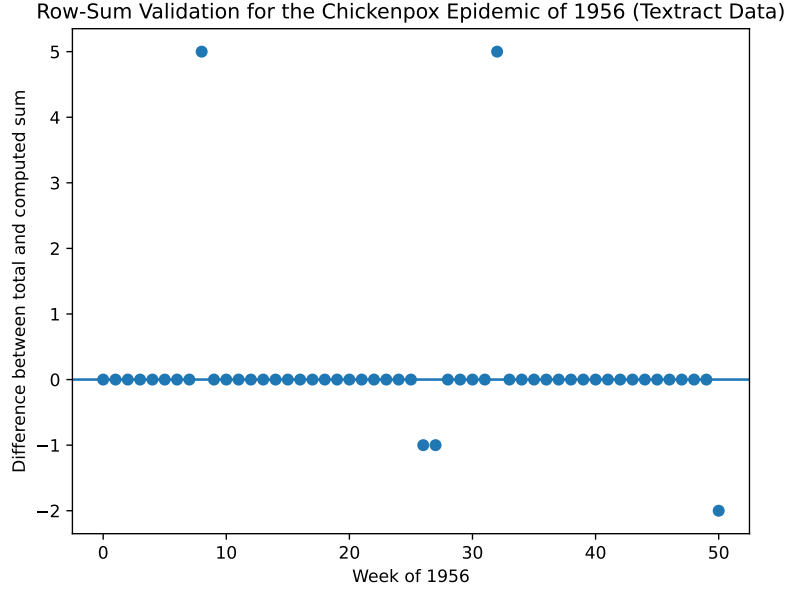


Figure 5: Differences in expected versus calculated row sums for Chickenpox in CANDID (section 3.1) to verify accuracy of cell values (method described in section 4).

inspecting the scan of the table, we find the true total is *1224*, computed from the non-empty contributing values:

$$55, 495, 379, 25, 270$$

In the extracted version of the dataset, Textract correctly extracts the total to be *1224* from the table, but differs from the reference values on the actual contributing incidents values:

$$50, 495, 379, 25, 270$$

Our algorithm found these values and calculated the sum, which is 1,219. So the difference between the expected sum and the calculated sum is 5, as seen in fig. 5. A direct comparison shows that the value 55 was misread as 50. In this way, errors in relatively minor numerical errors can be found and corrected.

Some discrepancies are more subtle. For instance, the first outlier in fig. 7 corresponds to Week 9. The recomputed total from the extracted contributing values is:

$$5, 13, 59, 782, 802, 61, 2, 47, 111$$

Textract reports a total of *1852*, while the sum of the contributing values equals *1882*. This gives the difference of -30 in fig. 7 at week 9. This inconsistency indicates that there exists an error from either the contributing cells or the extraction of the row total, or both. When we compared these values to the scanned document, we found that the reference transcription records the row total as *1862* (see fig. 6), which differs from Textract’s reported total of *1852* by only *10* – but a 10-count discrepancy between the two transcriptions does not explain the original 30-count gap between Textract’s total and the sum of the contributing values. A further gap of *20* (*1882* minus *1862*) separates the reference transcription’s total from the sum of the contributing values.

Upon closer inspection of the source image, the correct value in the scan of the total is *1882* (exactly the sum of the contributing values). This highlights an error in the reference transcription, not the Textract extraction, coming from the ambiguity of the original scan. This is a substantive result in its own right: row-sum validation detected an error in the reference transcription rather than in the Textract output, demonstrating that internal consistency checks can adjudicate between the two sources rather than merely score the extraction tool against an assumed-correct reference (see also the Discussion in section 6.2).

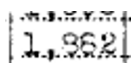


Figure 6: Total Measles instances in Week 9 of CANDID. This is hard to interpret for both OCR and humans.

These examples illustrate an important advantage of the row-sum validation approach: it detects small inconsistencies in numerical cell contents, and this approach can be adapted

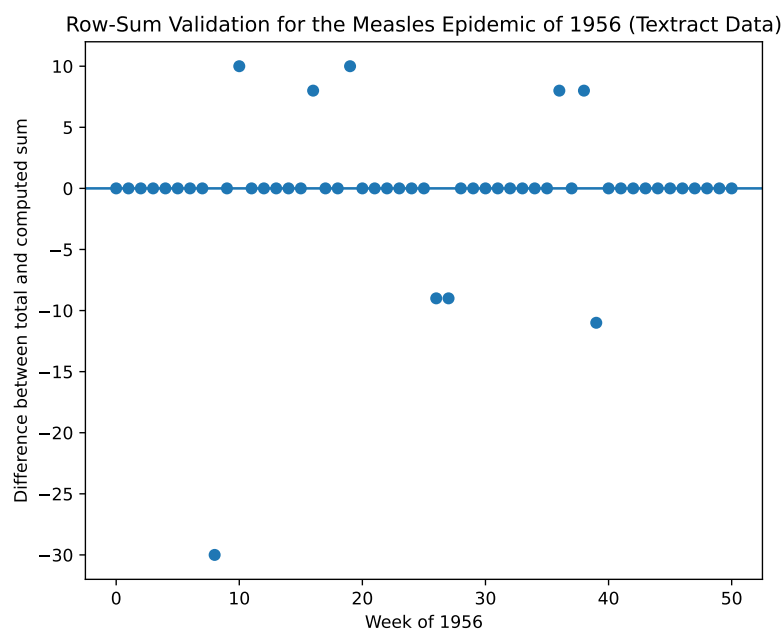


Figure 7: Differences in expected versus calculated row sums for Measles in CANDID (section 3.1) to verify accuracy of cell values (method described in section 4).

487 for the detection of OCR errors. By comparing independently computed totals with reported
 488 values, the method provides a mechanism for flagging anomalies in the dataset.

5.2 Comparing Important Statistics

5.2.1 Cross-Correlation Between Diseases

Unless otherwise stated, the Textract series used in this subsection are the processed data described in table 1 (Stage 2), without manual correction. The cross-correlation analysis presented first focuses on the relationship between Chickenpox and Measles epidemics. This involved extracting the relevant time series from both the manually curated dataset and the Textract extraction, followed by post-processing to align formats, handle missing values, and ensure consistent indexing across both sources (see section 2.2).

To provide an initial qualitative assessment of the relationship between these diseases, and to evaluate the accuracy of the extracted data, we plot the time-series of both datasets fig. 8. Visually, the two series are highly similar, indicating that the extraction process preserves the overall timing structure of the data.

This observation is consistent with the quantitative results reported in section 5.1.1, where it was established that content-only agreement (tolerant of formatting and of a value shifted into an adjacent column) is 95.6% for numeric-only cells, with most remaining errors involving only small numerical deviations. As a result, the dominant trends, seasonal patterns, and relative magnitudes of the two diseases are largely unaffected by extraction noise. The remaining disagreements do change the graphs slightly. At around week 60, there is a sharp decrease in Chickenpox, and around week 145, there is some Measles data missing. This is also unsurprising to see in the Textract data, which contains errors that are systematic and sharp; for example, the Scarlet Fever comma-induced shift described in section 5.3 propagates through all affected rows rather than occurring at random.

The two series rise and fall in roughly the same seasonal pattern, with major peaks and troughs occurring at approximately the same times. Thus, from the incidence curves alone, we would expect a fairly strong positive cross-correlation at lag 0. The cross-correlation function would likely show its maximum at or very near lag 0, with a value that is moderately

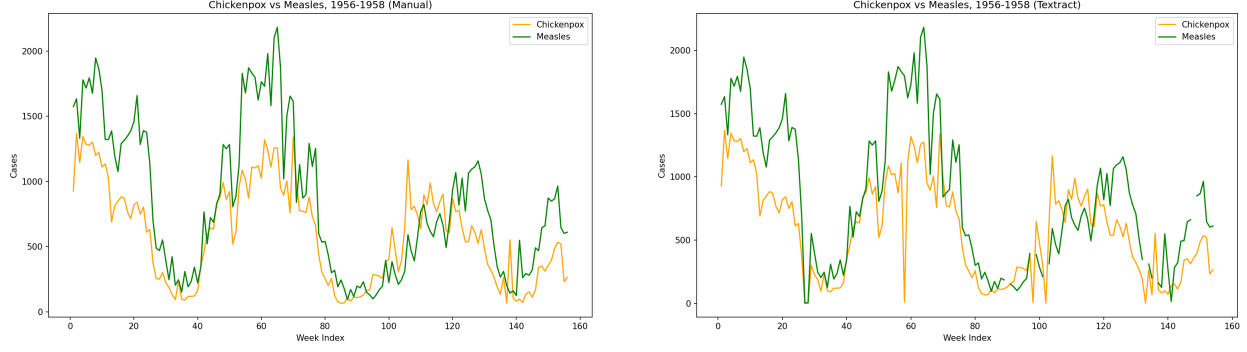


Figure 8: Comparison of manual versus Textract disease incidence for Chickenpox and Measles. Data version: Textract series processed through the pipeline in table 1 (Stage 2), without manual correction.

high after standardization.

For our purposes, it is critical to examine the difference between the cross-correlation computed between the manual and Textract data. The graphs for both computations are shown in fig. 9; the quantified comparison follows below (table 3).

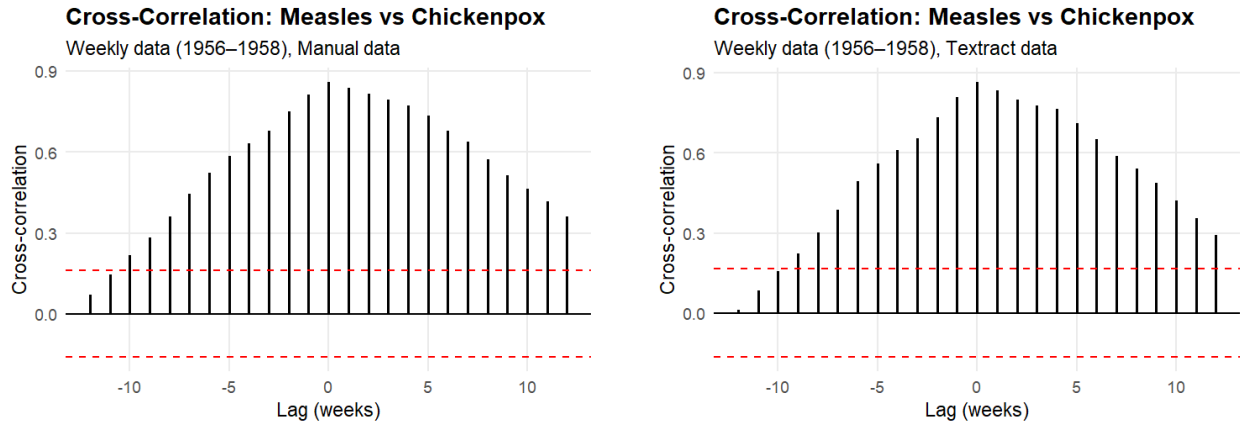


Figure 9: Comparison of manual versus Textract disease cross-correlation for Chickenpox and Measles. This is described in section 4. Data version: processed Textract data (table 1, Stage 2), without manual correction.

To gain a sense of how similar the cross-correlations are, we computed a set of summary statistics across all lags (table 3; per-lag values are in , Table S1). We report absolute and root-mean-square differences rather than a relative difference at each lag, because a relative difference is uninformative wherever the manual correlation is itself near zero, as the next comparison illustrates.

Both datasets peak at lag 0 (table 3), so the change in peak lag is 0 and the change in peak correlation is small (-0.0271). Across all 25 lags, the largest absolute discrepancy between the two series is 0.0832 (at lag 7) and the root-mean-square difference is 0.0526. Importantly, the lag with the highest cross-correlation remains unchanged, demonstrating that Textract does not distort the timing structure of the series. Because cross-correlation is a measure of time relationships rather than exact magnitudes, we see that the Textract data performs well.

We repeat this process with the time series for Measles and Meningitis (incidence curves in fig. 10, cross-correlation in fig. 12), and compare the values using the same summary statistics (table 3; per-lag values are in , Table S2). Unlike the Measles/Chickenpox comparison above, the Meningitis series used here has been manually corrected: additional post-processing was applied to remove visible outliers, as described in section 5.3.

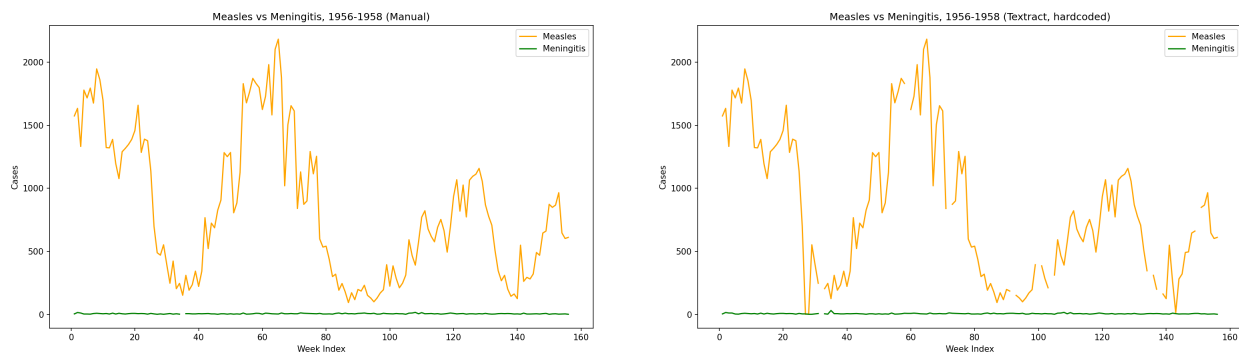


Figure 10: Comparison of manual versus Textract disease incidence for Measles and Meningitis. Data version: manually corrected (outlier removal, section 5.3). The Meningitis data without this additional step is seen in fig. 11.

The cross-correlation between Measles and Meningitis is weak across all lags in both the manual and Textract datasets, with values generally between about -0.11 and 0.09 (table 3). Because these correlations are close to zero throughout, this pair is the one for which a relative-difference metric was least informative: at several lags the manual correlation itself is close to zero, so any ratio against it is unstable regardless of how small the absolute discrepancy is. The manual series peaks at lag -9 and the Textract series at lag -11 , a

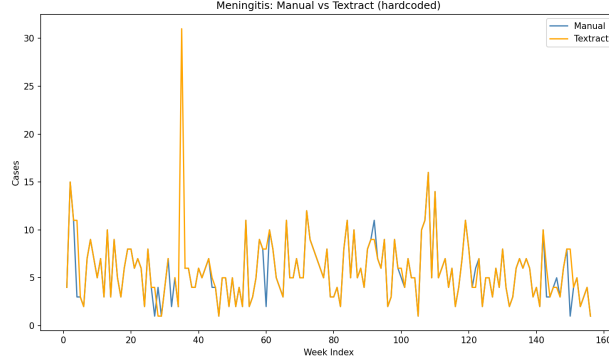


Figure 11: Manual (blue) and Textract (orange) time series for Meningitis. The Textract data shown here is plotted *after* the additional post-processing seen in section 5.3; see table 3 for the quantified comparison.

change in peak lag of -2 weeks; the change in peak correlation is small (-0.0059). The largest absolute discrepancy between the two series is 0.1108 (at lag -7 , where Textract crosses to a small negative value while the manual value remains small and positive) and the root-mean-square difference is 0.0423 – both smaller in absolute terms than the Measles/Chickenpox discrepancies above, even though a relative-difference expression of this same gap would read an order of magnitude larger, for the reason given above. Textract captures the overall cross-correlation pattern here, and the largest discrepancies occur at lags where both correlations are already small.

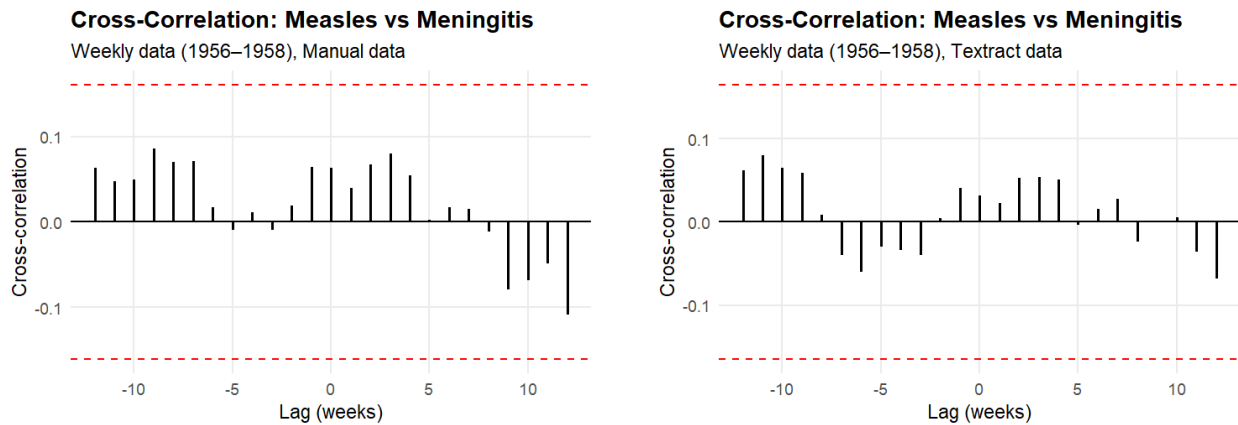


Figure 12: Comparison of manual versus Textract disease cross-correlation for Measles and Meningitis. Data version: manually corrected (outlier removal, section 5.3).

Finally, the same procedure is repeated for a third pair of diseases, namely Chicken-

551 pox and Mumps. The Textract series used here are manually corrected, following the same
 552 procedure applied to the Meningitis data (section 5.3). Time series for both diseases are
 553 extracted from the manually curated dataset, and the Textract output, and the correspond-
 554 ing incidence curves are plotted in fig. 13. Prior to computing the cross-correlation, the
 555 extracted data undergoes the same post-processing pipeline used for the Meningitis dataset.
 556 In particular, OCR-related errors identified as outliers are manually corrected to ensure that
 557 the resulting time series is suitable for analysis. This correction procedure is described in
 558 detail in section 5.3.

559 We see in fig. 14 that the graphs of the cross-correlation calculated between Chickenpox
 560 and Mumps are very similar between the manual and Textract data. The summary statistics
 561 in table 3 (per-lag values in , Table S3) confirm this: the differences between the two datasets
 562 at each lag are small.

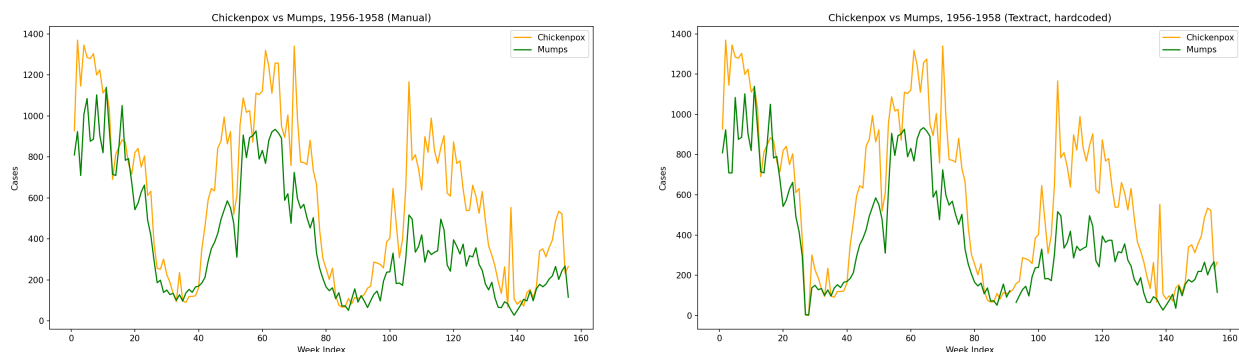


Figure 13: Comparison of manual versus Textract disease incidence for Chickenpox and Mumps. Data version: manually corrected (section 5.3).

563 The cross-correlation between diseases and the conclusions we draw from their CCF data
 564 are almost identical in both the manual and Textract dataset for this pair: the peak lag,
 565 which is most important for epidemiological study, is identical (lag 0 in both), the change in
 566 peak correlation is negligible (-0.0016), and both the largest absolute discrepancy (0.0205 ,
 567 at lag 12) and the root-mean-square difference (0.0106) are the smallest of the three disease
 568 pairs examined.

569 Table 3 collects these summary statistics for all three disease pairs. A strong cross-

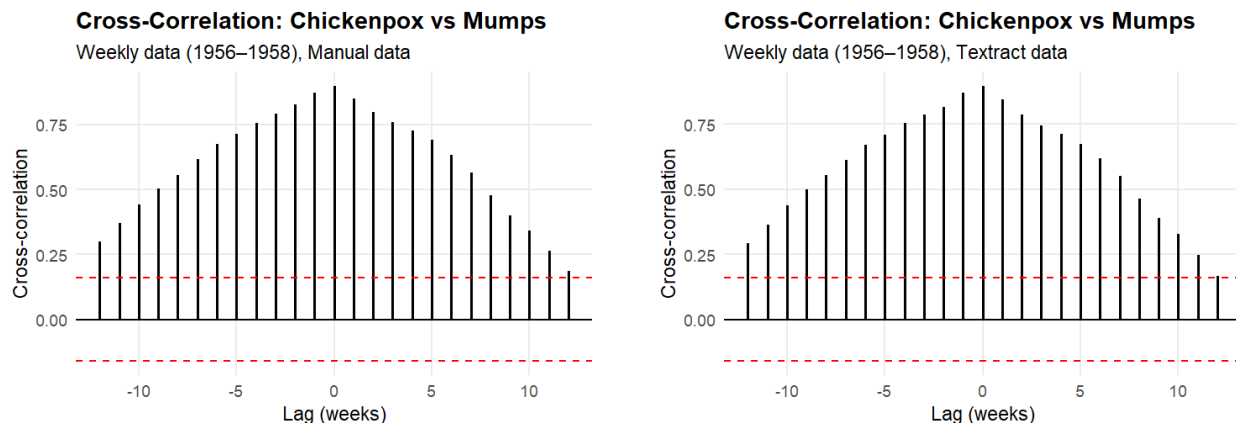


Figure 14: Comparison of manual versus Textract disease cross-correlation for Chickenpox and Mumps. Data version: manually corrected (section 5.3).

correlation near lag zero is expected for two seasonal childhood infections circulating in the same population and is not, on its own, a demanding validation target; what these comparisons test is whether Textract preserves the *structure* of that dependence (peak lag, and the overall shape of the CCF) rather than establishing a stringent benchmark of extraction accuracy.

Disease pair	Data version	Max abs diff	Lag at max	RMS diff	Peak lag M	Peak lag T
Measles vs Chickenpox	processed, no manual correction	0.0832	7	0.0526	0	0
Measles vs Meningitis	manually corrected	0.1108	−7	0.0423	−9	−11
Chickenpox vs Mumps	manually corrected	0.0205	12	0.0106	0	0

Table 3: Summary of cross-correlation agreement between manual and Textract series for all three disease pairs, computed by `analysis/ccf_summary.py` from the per-lag CCF values (, Tables S1–S3). Lag range $k \in \{-12, \dots, +12\}$ ($n = 25$ lags). Sign convention: all differences are Textract − Manual. “M”/“T” denote Manual/Textract. Peak-lag ties (S3.2: smaller $|k|$, then negative lag) did not occur for any of the three pairs. **Data version** follows table 1: two of the three pairs (Meningitis, Mumps) use manually adjudicated Textract series (section 5.3); Measles vs. Chickenpox does not.

5.2.2 Time Series Statistics

After isolating the time series for specific diseases in CANDID, we can find the peak heights and times. In fig. 8 we can see, in blue, the time series for Measles. As mentioned, the Textract data is very visually similar to the manually entered data.

We report both the initial exponential growth rate r and the doubling time T_d it implies for the Measles epidemics: r is the quantity actually estimated by the model, and T_d is a transformation of it that is easier to interpret numerically, so reporting T_d alone would hide the scale on which the model was fitted. Both are calculated with the R package `epigrowthfit` [18], using the model specification given in section 4. Because our incidence data are indexed in weeks, r is a per-week rate. The graphs below show the fitted curves and doubling-time annotation for Measles for both datasets (see fig. 15).

We see in table 4 the numeric results comparing growth rate, doubling time, peak week, and peak cases between the manually entered and automated datasets.

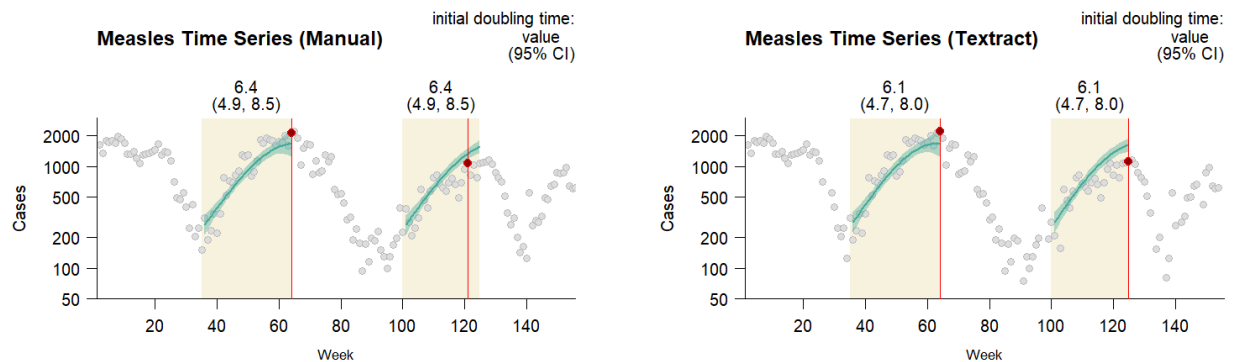


Figure 15: Manual versus Textract Measles growth rates as described in section 4. Data version: processed Textract data (table 1, Stage 2), without manual correction. Shaded bands mark the two fitting windows (week 35–64 and week 95–126). The annotated “initial doubling time: value (95% CI)” is in weeks and is the same number in both shaded windows for a given panel: this is not a labelling error but a direct consequence of the model specification (section 4), which estimates one growth rate shared across both windows rather than a separate rate per window; see table 4 for the corresponding growth rate r .

We can compare these same statistics with a different disease. For example, fig. 16

Window	Fit range (wk)	n obs	Source	Peak week	Peak cases	Δ week	Δ cases (%)	r (wk ⁻¹ , 95% CI)	T_d (wk, 95% CI)
Window 1	35–64	30	Manual	64	1967	–	–	0.137 (0.114, 0.165)	5.1 (4.2, 6.1)
		30	Textract	64	2161	0	+194 (+9.84)	0.124 (0.100, 0.154)	5.6 (4.5, 6.9)
Window 2	95–126	32	Manual	126	1040	–	–	0.137 (0.114, 0.165)	5.1 (4.2, 6.1)
		30	Textract	126	1174	0	+134 (+12.91)	0.124 (0.100, 0.154)	5.6 (4.5, 6.9)

Table 4: Comparison of peak week, peak cases, growth rate, and doubling time between Textract and manual data for Measles, recomputed from `data/analysis_intermediates/table4_measles_peak_comparison.csv` and `growth_rate_summary.csv` (analysis/growthrates.R). Data version: processed Textract data (table 1, Stage 2), without manual correction – the only Textract version available for Measles (see section 5). Sign convention: Δ = Textract – Manual. Peaks are located from a LOESS-smoothed series within each fit window (peak-detection rule, section 4); ties resolve to the earliest week. n obs is 30 for every window/source except Manual Window 2 (32); the Textract series is missing 2 weeks (weeks 98 and 102) inside Window 2, which is why the Manual and Textract fits for that window are not run on identical n . r and T_d are estimated from a single growth-rate parameter shared across both fitting windows (model specification, section 4) and are therefore identical for Window 1 and Window 2 within a given source; both fits converged (`optim` status “relative convergence”). The Manual/Textract 95% CIs for r overlap and each point estimate falls within the other source’s CI, so the two are not distinguishable at the precision these data support (section 5); this comparison is not on strictly identical n given the two missing Textract weeks noted above.

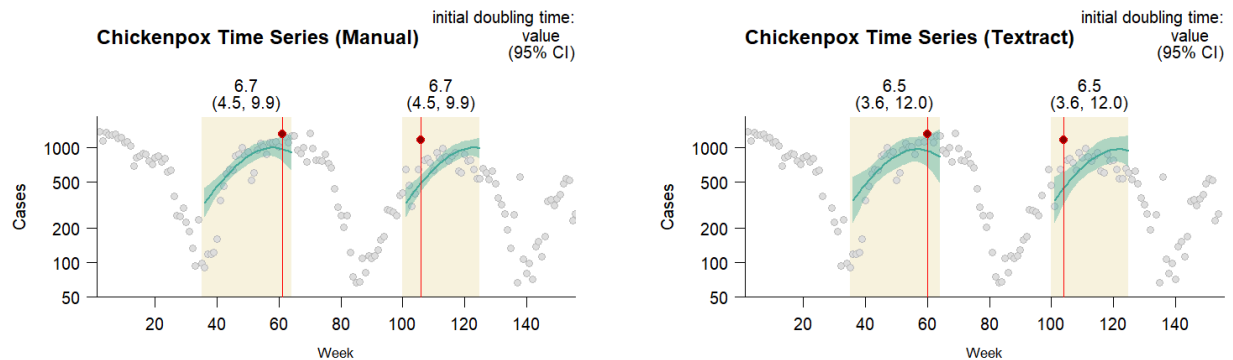


Figure 16: Manual versus Textract Chickenpox growth rates as described in section 4. Data version: processed Textract data (table 1, Stage 2), without manual correction. Shaded bands mark the two fitting windows (week 35–64 and week 95–126). The annotated “initial doubling time: value (95% CI)” is in weeks and is the same number in both shaded windows for a given panel: this is not a labelling error but a direct consequence of the model specification (section 4), which estimates one growth rate shared across both windows rather than a separate rate per window; see table 5 for the corresponding growth rate r .

Window	Fit range (wk)	n obs	Source	Peak week	Peak cases	Δ week	Δ cases (%)	r (wk ⁻¹ , 95% CI)	T_d (wk, 95% CI)
Window 1	35–64	30	Manual	64	1223	–	–	0.144 (0.125, 0.167)	4.8 (4.2, 5.5)
		30	Textract	64	1313	0	+90 (+7.42)	0.144 (0.108, 0.193)	4.8 (3.6, 6.4)
Window 2	95–126	32	Manual	114	882	–	–	0.144 (0.125, 0.167)	4.8 (4.2, 5.5)
		32	Textract	112	882	–2	0 (0.00)	0.144 (0.108, 0.193)	4.8 (3.6, 6.4)

Table 5: Comparison of peak week, peak cases, growth rate, and doubling time for Chickenpox between Textract and manual data, recomputed from `data/analysis_intermediates/table5_chickenpox_peak_comparison.csv` and `growth_rate_summary.csv` (analysis/growthrates.R). Data version: processed Textract data (table 1, Stage 2), without manual correction – the only Textract version available for Chickenpox (see section 5). Sign convention: Δ = Textract – Manual. Peaks are located from a LOESS-smoothed series within each fit window (peak-detection rule, section 4); ties resolve to the earliest week. r and T_d are estimated from a single growth-rate parameter shared across both fitting windows and are therefore identical for Window 1 and Window 2 within a given source; both fits converged. The Manual/Textract 95% CIs for r overlap and each point estimate falls within the other source’s CI, so the two are not distinguishable at the precision these data support (section 5); note the Textract CI is noticeably wider than the Manual CI (0.108–0.193 vs. 0.125–0.167).

showcases the doubling times and peaks for Chickenpox as seen in the manual and extracted versions of the dataset. These statistics are compared in Table 5.

Peak week is unchanged (Textract – Manual = 0) in three of the four windows, and differs by –2 weeks in the fourth (Chickenpox, Window 2); peak case counts differ by up to +194 cases (Measles Window 1, +9.84%) in absolute terms and by up to +12.91% (Measles Window 2, +134 cases) in relative terms – the largest absolute and largest relative discrepancies fall in different windows – with Textract consistently over-counting relative to Manual across all four windows. None of the four reported peak weeks can be cross-checked against the row-sum validation of section 5.3, since that validation covers only 1956 while all four peaks fall in 1957–1958 (section 4).

For the growth rate itself: the Measles estimate is $r = 0.137$ wk⁻¹ (95% CI 0.114–0.165) from the manual series and $r = 0.124$ wk⁻¹ (95% CI 0.100–0.154) from Textract, a difference of $\Delta r = -0.013$ wk⁻¹; the corresponding doubling times are 5.1 (4.2, 6.1) and 5.6 (4.5, 6.9) weeks, $\Delta T_d = +0.5$ weeks. For Chickenpox, $r = 0.144$ wk⁻¹ from both sources to three significant figures (95% CI 0.125–0.167 manual, 0.108–0.193 Textract), $\Delta r \approx -0.0004$ wk⁻¹; doubling times are 4.8 (4.2, 5.5) and 4.8 (3.6, 6.4) weeks, $\Delta T_d = +0.01$ weeks. In both

diseases the Manual and Textract 95% CIs for r overlap, and each point estimate falls inside the other source’s interval; the Chickenpox Textract CI is also visibly wider than its Manual counterpart, which is expected given no more data points were available to the fit (tables 4 to 5 report n observations per window). Rather than describing these as “nearly identical,” the defensible statement is one of indistinguishability given the precision the data support: the Textract-based growth-rate and doubling-time estimates cannot be distinguished from the manual-based estimates at the resolution of these fits, for either disease or either window tested. This is a weaker claim than exact agreement, and it is scoped to the two diseases (Measles, Chickenpox) and two epidemic windows examined here; it does not establish that growth-rate estimation from Textract data is reliable in general.

5.3 Remarks on Dataset Specific Coding

Several challenges arose when computing row-wise sums for each disease. The primary issue was the extraction errors, such as row misalignment or mislabeled diseases. These issues shifted entire rows of values and consistently propagated across tables, making it difficult to associate values with the correct disease directly. For example, if a disease name contains a comma in the original table, Textract extracts this as two separate cells, shifting the following disease incidence values. As each table has the same or similar disease names, this problem in extraction appeared in each table corresponding to that row.

Grouping entries by disease (rather than by week as extracted) before performing the summation reduced this complexity. This limited the need for extensive manual intervention, as most diseases had identical errors in each table.

When a shift occurred (typically due to errors in disease labels) the position of the reported total and its corresponding contributing values became offset, and the indices used to identify these values were no longer valid. To address this, the correct indices were re-identified through manual inspection of the extracted data for each disease, and the summation inputs were adjusted accordingly.

631 An example of this shifting in CANDID occurs with Scarlet Fever. Persistent discrep-
632 ancies were observed between reported and recomputed sums. Beginning in week 5 of 1956,
633 all row sums appeared incorrect. Investigation revealed that the disease label had changed
634 from “Scarlet Fever” to “Scarlet Fever, etc.” in the original document. The introduction of
635 a comma altered the table parsing, causing a shift in all subsequent cell values within the
636 row. This shift propagated through the extraction, leading to misalignment. Resolving this
637 issue required manually identifying the change and recombining the affected cell values to
638 restore the correct structure.

639 A further consideration in the row-sum computation is the structure of the original tables:
640 a row index is printed both at the beginning of each row and repeated at the end (see fig. 1),
641 and the province “Report Week” values are not the only numeric-looking entries in the
642 row. Rather than iterating generically across the row and attempting to filter out non-
643 contributing values at runtime, the row-sum script (`analysis/row_sum_check.py`) avoids
644 this problem by construction: it reads the Canada report-week total from a fixed column
645 and sums only the fixed set of columns, identified directly from the table layout in fig. 1,
646 that hold each province’s report-week value (fig. 17). Because the contributing columns
647 are specified explicitly rather than inferred from cell content, the row index and any other
648 non-contributing cell are never candidates for inclusion in the sum.

```
TARGET_COL = 2
CONTRIB_COLS = list(range(8, 36, 3)) # 8, 11, 14, ..., 35 -- one
    per province

target = to_int_or_none(row[TARGET_COL])
contributors = [(c, to_int_or_none(row[c])) for c in CONTRIB_COLS]
computed = sum(v for _, v in contributors if v is not None)
```

Figure 17: The column layout used by the row-sum validation script (`analysis/row_sum_check.py`) to identify the Canada report-week total (column 2) and each province’s contributing report-week value (columns 8, 11, 14, ..., 35, one per province at three-column intervals), as reproduced from the actual script rather than paraphrased.

We also had to correct extraction errors when comparing the epidemiological statistics between the two datasets. The extracted Meningitis time series contains errors that appear as large, isolated spikes, which are not consistent with the expected disease patterns. Because these errors would distort the results of the cross-correlation analysis, they were corrected before proceeding. The unprocessed time series for Measles and Meningitis is shown in fig. 19. The Measles data follows a smooth and expected pattern, while the Meningitis data includes several sharp spikes that do not match the original dataset. These spikes are caused by OCR errors, where some values are misread or incorrectly extracted.

The following is another example of the Textract time series for Mumps versus Chickenpox, before our manual correction (see fig. 18).

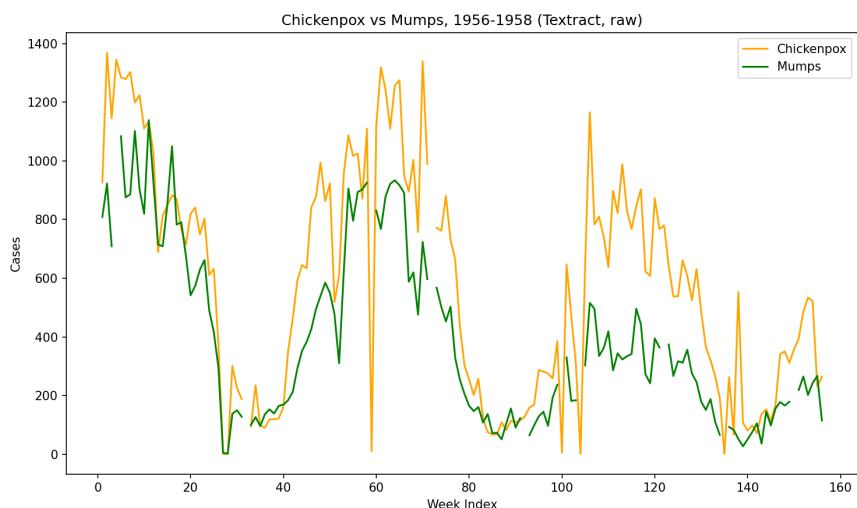


Figure 18: Data version: raw/uncorrected Textract values (prior to the manual correction in section 5.3). The time series for Chickenpox and Mumps from Textract, with clear errors seen around weeks 60, 100, 110, etc as well as missing values.

Although these errors are large, they occur only occasionally: a small number of spikes were identified across the affected time series. We identified them using a boxplot or violin plot (see fig. 20), which highlights these extreme values compared to the rest of the data. This method identified the extreme values in our data.

Correcting these incidence values by checking against the original dataset took less than 20 minutes for the Meningitis and Mumps series reviewed here (three years of weekly data).

665 This is a measured correction time for these specific series, not an estimate of total post-
666 processing effort across the full dataset.

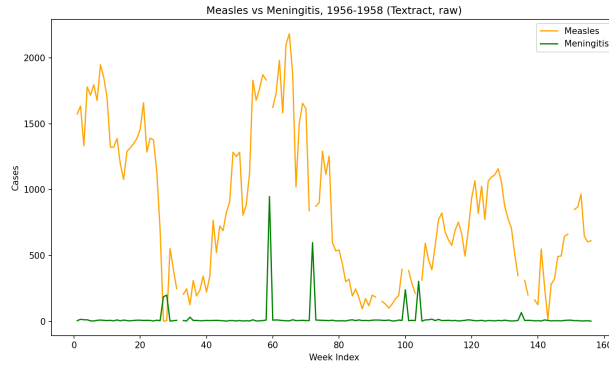


Figure 19: Data version: raw/uncorrected Textract values (prior to the manual correction in section 5.3), showing clear errors in the meningitis cases.

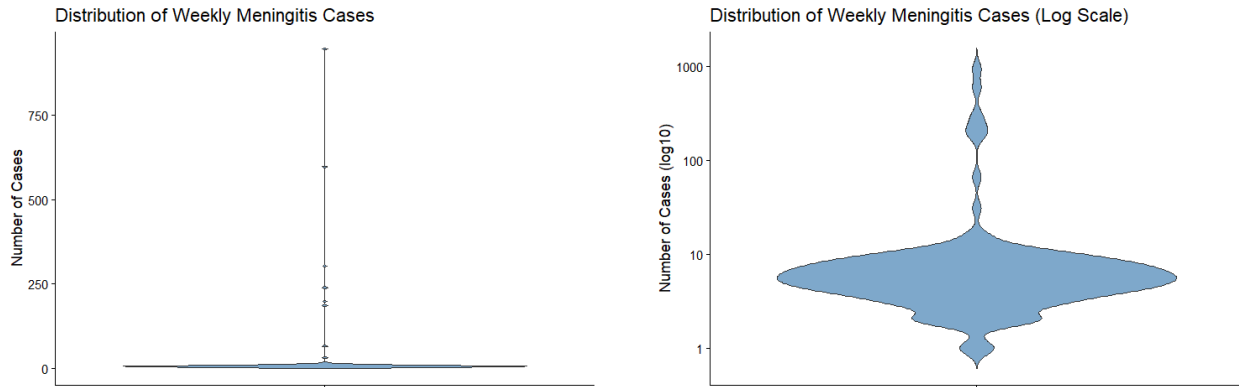


Figure 20: Violin plots of Textract Meningitis data shown on linear (left) and log (right) scales.

6 Conclusion

6.1 Impact

From the perspective of epidemiology researchers, automating tabular data entry is a major aid in studying historical data. It accelerates the processing of large datasets and frees effort and resources for interpretation rather than manual transcription. Because historical sources are often vast, messy, and inconsistent, automation makes it possible to standardize and analyze them at scale, revealing patterns and insights that would have been implausible to uncover in the same amount of time with only manual data entry. The conclusions in this paper are limited to printed, regularly formatted CANDID tables from 1956–1958 processed by the specific extraction and post-processing workflow described above; we do not claim these results generalize to handwritten documents, irregularly formatted tables, or other data-extraction tools or model versions.

For an automation tool to successfully improve and accelerate the data entry process, it must meet all or most of the requirements we have set in this paper. That is, the software must be relatively simple to use and produce a dataset that is either

1. Close to the original dataset or
2. Easy to process and fix the errors it produces

If too much human correction is necessary, the time saved becomes relatively small, so the tool is not successful. As well, even with modest time saving, data correction is a less pleasant task for humans than data entry. In our own preliminary exploration (section 1), the conversational model we tried before Textract produced errors prevalent and tedious enough to fix that we abandoned it in favour of a tool purpose-built for table extraction.

In comparison, the errors present in Textract’s table extraction are shown to be of a different character than typical manual entry errors. Textract’s errors are often either small

(differing from the reference value by one or two characters, with a small influence on various statistical tests on the dataset) or large. This second, large-magnitude type of error is relatively rare and occurs specifically in the extraction of disease-name string values (section 4.1.2).

Beyond the initial technical hurdle of utilizing AWS Textract programmatically, as well as the unique post-processing necessary for each dataset, the overall time and effort necessary to recreate CANDID with Textract appears to be reduced in comparison to manual data entry. Similar tables (in both size and complexity) took on average 30 minutes to fully digitize from a scanned copy by a member of McMaster’s Theoretical Biology group; extrapolated at that single operator’s rate to all 156 tables from 1956 to 1958, manual data entry would take an estimated 78 hours. This is not a measurement of the full manual process, but an extrapolation from one operator’s rate on comparable tables. On the automated side, the components we directly measured are: script runtime (≈ 15 minutes) and the correction pass (see section 5.3, < 20 minutes for the reviewed series). Script development time and the remaining post-processing time were not measured. With these caveats, we conclude that for similar tables, extracting data with Textract substantially reduced estimated labour in the data entry process and can be supplemented with dataset-specific post-processing.

6.2 Limitations

(a) No adjudicated reference standard. The manual transcription is used throughout this paper as the comparison basis without independent re-adjudication: there was no double transcription and no two-reviewer resolution of discordant cells. The paper itself demonstrates that this reference can be wrong – the Week 9 Measles total (section 5), where the source scan supports *1882* and the reference transcription records *1862*. Reported disagreement counts therefore conflate extraction errors with transcription errors, and the direction of this bias is unknown: some fraction of the cells we count as Textract errors may instead be reference-transcription errors, and vice versa. Resolving this would require two

independent reviewers adjudicating a random validation sample against the source scans, with inter-reviewer agreement documented; this was not done for this paper.

A further limitation concerns data leakage in the development of our normalization and structural-repair rules. The comma/whitespace normalization, header-exclusion, and row-shift realignment rules described in section 5.1.1 (see table 1) were developed, tuned, or validated while the reference transcription was visible to us; no held-out test set of unseen tables was used. Specific instances of this include the row-shift realignment rule (section 2.2), the Scarlet Fever comma fix (section 5.3), the fixed-column row-sum computation that avoids the repeated row index by construction (section 5.3), and the Meningitis/Mumps spike correction (section 5.3). Because these rules were shaped by the specific errors we observed in this dataset, the reported agreement figures should be read as an upper bound on the performance we would expect on previously unseen CANDID tables or other datasets, rather than as an estimate of out-of-sample accuracy.

(d) Evaluation year. Cell-level agreement (section 5.1.1–section 4.1.2) covers 1956 only (52 of the 156 tables); the downstream epidemiological statistics (section 5) use all three years, 1956–1958. This is stated where the relevant numbers are first reported, but is worth naming explicitly as a limitation: the peak weeks reported in the growth-rate tables (weeks 64, 112, 114, and 126) sit entirely outside the range for which cell-level agreement was directly measured, so their accuracy rests on the assumption – untested here – that agreement in 1957–1958 resembles agreement in 1956.

(f) Selection of downstream comparators. Three disease pairs for cross-correlation and two diseases for growth-rate comparison were chosen without a pre-specified selection rule. A strong cross-correlation near lag zero is expected for two seasonal childhood infections circulating in the same population and is not, on its own, a demanding validation target; what these comparisons test is whether Textract preserves the *structure* of that dependence rather than establishing a stringent benchmark of extraction accuracy.

(c) No external validation on a structurally different archive. All results in this paper come from a single source (CANDID) with a consistent printed template; transfer of these findings to other table layouts, languages, typographies, or scan qualities is untested. section 6.3 discusses a structurally distinct Quebec provincial dataset as a candidate for such a test; we have not analysed it here.

(e) Single tool, single version. This study evaluates one extraction tool and does not constitute a controlled comparison against alternatives; the informal exploration described in the Introduction that led us to Textract was not a rigorous benchmark, and no controlled comparison against other document-extraction tools or current document-capable language models was performed. Beyond that, we used one commercial service at one point in time, without pinning or recording a specific model or API version; Amazon may change Textract’s underlying models at any time without notice, so the extraction results reported here may not reproduce exactly if the tables were re-extracted today. Our choice of Textract reflects its suitability for this task rather than a demonstrated advantage over alternatives.

The growth-rate and doubling-time comparison (tables 4 to 5) is similarly limited in scope: only a single Textract data version – the Stage-2 processed series, without manual spike correction – is available for Measles and Chickenpox, since the manual correction procedure of section 5.3 was applied only to the Meningitis and Mumps series. We are therefore unable to test whether the growth-rate estimate is stable across raw, processed, and manually corrected versions of the same disease’s data, which would be a useful further check; doing so would require applying the same manual review used for Meningitis and Mumps to the Measles and Chickenpox series as well, which was not done for this paper.

(b) No measured time-and-cost study. The labour comparison in section 6.1 is an extrapolation from one operator’s rate on comparable tables, not a controlled measurement: script development time, dataset-specific coding time, validation time, retries, and AWS charges were not measured, and only the extraction runtime (≈ 15 minutes) and one cor-

reception pass (< 20 minutes, two series) were timed directly. A defensible comparison would record person-hours by activity, operator count and experience, machine time, itemised AWS charges with a pricing date, and would separate the first-project setup cost from the marginal cost of digitizing an additional year of tables.

The limitations above concern the validity and scope of what was measured. The remainder of this section concerns the operational cost of using Textract, which does not affect the validity of the results but is relevant to a researcher deciding whether to adopt this workflow.

The main operational limitation of Textract is the barrier to entry associated with AWS software. Limited experience with software development might increase the effort required to use Textract. Beyond account set-up and writing the extraction script, connecting the code to an AWS account involves an AWS ID, secret key, secret access key, and more (, S1). However, once the initial setup is complete, researchers have complete control over how the data is processed and extracted: for example, a researcher can choose to omit confidence levels, or process only certain elements of the table or certain pages of the dataset. Unlike the data-specific post-processing steps that follow extraction, the same extraction script can be used for any scanned document of a table.

Another operational limitation is the time required by the software to extract the raw data: the code to extract the full CANDID dataset of 156 tables took around 15 minutes to run and terminate. This could likely be reduced, for instance by cropping the original scans to the relevant part of the table or by omitting confidence levels, but optimizing the API calls was not a priority of this project.

The largest operational limitation is the time and effort required to process and clean the data post-extraction: formatting errors such as merged-cell errors and shifted row values, and the large errors Textract produces in rare numeric values and string headers, together took more time to find and correct than running the extraction code did. Small errors were present throughout our extraction and can be particularly difficult to detect, especially when they occur in numerical values that appear plausible on their own. These errors may arise

from OCR misreads, formatting inconsistencies, or noise in the original documents, and they often do not stand out unless carefully validated against source data.

6.3 Future Work

Future trials on different datasets will reveal more about the true power and limitations of tools such as Textract. We can expand the scope of our trials to more complicated tables, and to handwritten documents that contain infectious disease data. One natural next test case is a Quebec provincial communicable-disease dataset (Dominion Bureau of Statistics, July 1927–December 1931), collected and organized by McMaster’s Theoretical Biology group but not previously digitized or manually transcribed. It is structurally distinct from CANDID in ways that would stress-test the workflow described here: row headers are printed on the diagonal and in French, and the table layout changes partway through the series from two tables per page to one. We have not analysed this dataset in the present study – doing so properly would require an independent manual transcription of a stratified sample, which is beyond the scope of this paper – but it is a natural candidate for validating the workflow’s generalizability beyond a single archive.

Our current results suggest that, even when the initial extraction performs reasonably well on suitable datasets, the bulk of human effort is still concentrated in post-extraction data processing. This includes cleaning inconsistencies, correcting misread values, aligning rows and columns, and validating outputs against source documents. In other words, the main bottleneck is not the extraction itself, but the downstream effort required to make the data usable for analysis.

Going forward, a key objective will be to determine how much of this post-processing stage can be automated. There is strong potential for machine learning tools to play a larger role here. For instance, errors in natural language, such as mislabeled row and column headers that lead to high Levenshtein distances, could be identified and corrected using LLMs trained on domain-specific terminology. Similarly, more sophisticated validation software could be

822 developed to flag anomalies automatically, such as sudden outliers or structural irregularities
823 in tables. These systems could either correct errors directly or generate targeted reports for
824 human review, thereby significantly reducing manual workload while improving consistency
825 and scalability.

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References

1. Zhong X, ShafieiBavani E, and Jimeno Yepes A. Image-based table recognition: data, model, and evaluation. *Computer Vision – ECCV 2020*. Springer, 2020 :564–80. DOI: 10.1007/978-3-030-58589-1_34. Available from: https://doi.org/10.1007/978-3-030-58589-1_34
2. Hu J, Kashi RS, Lopresti D, and Wilfong GT. Evaluating the performance of table processing algorithms. *International Journal on Document Analysis and Recognition* 2002; 4:140–53. DOI: 10.1007/s100320200074. Available from: <https://doi.org/10.1007/s100320200074>
3. Mori S, Suen CY, and Yamamoto K. Historical review of OCR research and development. *Proceedings of the IEEE* 1992; 80:1029–58. DOI: 10.1109/5.156468. Available from: <https://doi.org/10.1109/5.156468>
4. Smith R. An Overview of the Tesseract OCR Engine. *Proceedings of the Ninth International Conference on Document Analysis and Recognition (ICDAR 2007)*. No DOI found. Washington, DC, USA: IEEE Computer Society, 2007 :629–33. Available from: <https://storage.googleapis.com/pub-tools-public-publication-data/pdf/33418.pdf>
5. Xia S, Xiong J, Dong H, Zhao J, Tian Y, Zhou M, He Y, Han S, and Zhang D. Vision language models for spreadsheet understanding: Challenges and opportunities. *arXiv preprint* 2024. DOI: 10.48550/arXiv.2405.16234. arXiv: 2405.16234 [cs.CV]. Available from: <https://doi.org/10.48550/arXiv.2405.16234>
6. Sahasrabudhe C, Ho Y, Winovich N, and Rajamanickam S. Imperfect Recognition: A Study of OCR Limitations in the Context of Scientific Documents. *2025 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. 2025

:996–1002. DOI: 10.1109/IPDPSW66978.2025.00155. Available from: <https://doi.org/10.1109/IPDPSW66978.2025.00155>

7. Panhuis WG van, Grefenstette J, Jung SY, Chok NS, Cross A, Eng H, Lee BY, Zadorozhny V, Brown S, Cummings D, and Burke DS. Contagious diseases in the United States from 1888 to the present. *New England Journal of Medicine* 2013; 369:2152–8. DOI: 10.1056/NEJMms1215400. Available from: <https://doi.org/10.1056/NEJMms1215400>
8. Panhuis WG van, Cross A, and Burke DS. Project Tycho 2.0: a repository to improve the integration and reuse of data for global population health. *Journal of the American Medical Informatics Association* 2018; 25:1608–17. DOI: 10.1093/jamia/ocy123. Available from: <https://doi.org/10.1093/jamia/ocy123>
9. Sockett PN, Garnett MJ, and Scott C. Communicable disease surveillance: Notification of infectious diseases in Canada. *Canadian Journal of Infectious Diseases* 1996; 7:293–5. DOI: 10.1155/1996/279482. Available from: <https://doi.org/10.1155/1996/279482>
10. Totten S, Medaglia A, and McDermott S. Updates to Canadian Notifiable Disease Surveillance System. *Canada Communicable Disease Report* 2019; 45:257–61. DOI: 10.14745/ccdr.v45i10a02. Available from: <https://doi.org/10.14745/ccdr.v45i10a02>
11. Earn DJD, MacKinnon G, Manzin S, Roswell M, Cygu S, Shi C, Bolker BM, Dushoff J, and Walker SC. A century of weekly notifiable disease incidence data by province in Canada. *PLOS Global Public Health* 2025 Dec; 5:1–17. DOI: 10.1371/journal.pgph.0005550. Available from: <https://doi.org/10.1371/journal.pgph.0005550>
12. Stadler M, Bannert M, and Sailer M. Cognitive ease at a cost: LLMs reduce mental effort but compromise depth in student scientific inquiry. *Computers in Human Behavior*

2024; 160:108386. DOI: 10.1016/j.chb.2024.108386. Available from: <https://doi.org/10.1016/j.chb.2024.108386>

13. Amazon Web Services. OCR software, data extraction tool — Amazon Textract. 2025. Available from: <https://aws.amazon.com/textract/>

14. Levenshtein VI. Binary codes capable of correcting deletions, insertions, and reversals. Soviet Physics Doklady 1965; 10:707–10. Available from: <https://api.semanticscholar.org/CorpusID:60827152>

15. Wagner RA and Fischer MJ. The String-to-String Correction Problem. Journal of the ACM 1974; 21:168–73. DOI: 10.1145/321796.321811

16. Wallinga J and Lipsitch M. How generation intervals shape the relationship between growth rates and reproductive numbers. Proceedings of the Royal Society B: Biological Sciences 2007; 274:599–604. DOI: 10.1098/rspb.2006.3754. Available from: <https://doi.org/10.1098/rspb.2006.3754>

17. Ma J, Dushoff J, Bolker BM, and Earn DJD. Estimating initial epidemic growth rates. Bulletin of Mathematical Biology 2014; 76:245–60. DOI: 10.1007/s11538-013-9918-2. Available from: <https://doi.org/10.1007/s11538-013-9918-2>

18. Jagan M and Bolker BM. epigrowthfit: Nonlinear Mixed Effects Models of Epidemic Growth. R package version 0.15.4. 2025. DOI: 10.32614/CRAN.package.epigrowthfit. Available from: <https://CRAN.R-project.org/package=epigrowthfit>

19. Amazon Web Services. Amazon Textract Developer Guide. Amazon Web Services Documentation. 2025. Available from: <https://docs.aws.amazon.com/textract/latest/dg/what-is.html> [Accessed on: 2025 Sep 5]

Supplementary Material

S1 Text. AWS Textract extraction script and setup instructions.

To run `AWS_Python_Table_Extraction.py`, first create a free AWS account at aws.amazon.com and sign in to the AWS Management Console. Go to the IAM service, create a new user (do not use root credentials), and attach a scoped inline policy granting only the action the script calls – `textract:AnalyzeDocument` – rather than the broad `AmazonTextractFullAccess` managed policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "textract:AnalyzeDocument",
      "Resource": "*"
    }
  ]
}
```

This scope was verified against the script's only Textract call, `client.analyze_document(...)`; if you extend the script to call other Textract operations (e.g. asynchronous jobs or `DetectDocumentText`), add the corresponding actions.

Generate an Access Key ID and Secret Access Key under that user's security credentials – save these immediately, as the secret key is shown only once. Install the AWS CLI and run `aws configure --profile textract-project` to store these credentials under a named profile rather than the default profile, entering the access key, secret key, and a default region (e.g., `us-east-1`, which matches the script's default). Set `AWS_PROFILE=textract-project` (or pass `profile_name` to `boto3.Session`) so the script authenticates as this scoped user; where your institution supports it, short-lived credentials issued via AWS IAM Identity

Center are preferable to long-lived access keys. Never commit credentials to version control, and rotate or delete the access key when the project ends. A free tier exists for Textract usage, with terms that vary by API operation and change over time; consult the current AWS Textract pricing page for the applicable limits and per-page rates before running a large extraction job.

Once credentials are configured, place your PDF or image files (PDF, PNG, JPG, or TIFF) in a single input folder and run the script from the command line as `python AWS_Python_Table_Extractor.py <input_dir> <output_dir>`, which will create two CSV files per document – one with extracted table cell text and one with per-cell confidence scores – in the specified output folder. Because this script uses Textract’s synchronous `AnalyzeDocument` API, each input file is limited to a single page; multi-page PDFs should be split beforehand (e.g., with a tool like `pypdf`) or processed via Textract’s asynchronous API if full multi-page support is needed. See also [19].

S2 Text. Repository.

For repository containing scripts and figures used see our repository.

S3 Text. Textract JSON block structure.

At this point in the API call (section 2), Textract does not yet know “rows” or “columns”; it only sees shapes and text. Textract returns a flat list of `BLOCK` objects (not a table as recognized by humans). The response is one JavaScript Object Notation (JSON) array called `Blocks`. Each block contains, among other information, `BlockType`, which contains values such as `TABLE`, `CELL`, `WORD`, and `SELECTION_ELEMENT`. Each block also contains a confidence score.

A `BlockType` of “TABLE” represents one detected table and contains a list of cell ids. Each `CELL` block contains `RowIndex`, `ColumnIndex`, `RowSpan/ColumnSpan`, and relationships to `WORD` blocks, which contain the actual string value of the cell along with its own confidence value.

S4 Text. Levenshtein distance recurrence.

Let $s = s_1 s_2 \dots s_m$ and $t = t_1 t_2 \dots t_n$ be two strings of lengths m and n . The Levenshtein distance $D(i, j)$ between the characters $s_1 \dots s_i$ and $t_1 \dots t_j$ is defined by the standard dynamic-programming recurrence [15]:

$$D(0, j) = j \quad \text{for } j = 0, 1, \dots, n$$

$$D(i, 0) = i \quad \text{for } i = 0, 1, \dots, m$$

$$D(i, j) = \min \begin{cases} D(i-1, j) + 1 & \text{(deletion)} \\ D(i, j-1) + 1 & \text{(insertion)} \\ D(i-1, j-1) + \mathbf{1}_{s_i \neq t_j} & \text{(substitution)} \end{cases}$$

$$\mathbf{1}_{s_i \neq t_j} = \begin{cases} 0, & s_i = t_j \\ 1, & s_i \neq t_j \end{cases}$$

The Levenshtein distance between strings s and t is then $\text{Lev}(s, t) = D(m, n)$.

S1 Table. Per-lag cross-correlation values, Measles vs. Chickenpox. Sign convention: Difference = Textract – Manual. Data version: processed, no manual correction (table 1). Summarized in table 3.

S2 Table. Per-lag cross-correlation values, Measles vs. Meningitis. Sign convention: Difference = Textract – Manual. Data version: manually corrected (section 5.3). Summarized in table 3.

S3 Table. Per-lag cross-correlation values, Chickenpox vs. Mumps. Sign convention: Difference = Textract – Manual. Data version: manually corrected (section 5.3). Summarized in table 3.

Lag	Manual	Texttract	Difference
-12	0.0710	0.0244	-0.0467
-11	0.1472	0.0871	-0.0601
-10	0.2187	0.1571	-0.0616
-9	0.2841	0.2199	-0.0642
-8	0.3621	0.3213	-0.0408
-7	0.4439	0.3996	-0.0443
-6	0.5214	0.4891	-0.0323
-5	0.5862	0.5315	-0.0548
-4	0.6308	0.5877	-0.0431
-3	0.6781	0.6264	-0.0518
-2	0.7497	0.7009	-0.0488
-1	0.8128	0.7766	-0.0362
0	0.8600	0.8329	-0.0271
1	0.8380	0.8078	-0.0301
2	0.8146	0.7682	-0.0464
3	0.7926	0.7385	-0.0542
4	0.7705	0.7412	-0.0292
5	0.7337	0.6668	-0.0669
6	0.6788	0.6102	-0.0686
7	0.6369	0.5536	-0.0832
8	0.5737	0.5395	-0.0341
9	0.5136	0.4662	-0.0474
10	0.4622	0.3991	-0.0631
11	0.4155	0.3386	-0.0769
12	0.3608	0.3091	-0.0516

Lag	Manual	Textextract	Difference
-12	0.0630	0.0620	-0.0010
-11	0.0472	0.0803	0.0331
-10	0.0495	0.0648	0.0153
-9	0.0862	0.0593	-0.0269
-8	0.0706	0.0086	-0.0620
-7	0.0707	-0.0402	-0.1108
-6	0.0175	-0.0603	-0.0778
-5	-0.0096	-0.0299	-0.0203
-4	0.0108	-0.0341	-0.0449
-3	-0.0092	-0.0395	-0.0303
-2	0.0195	0.0041	-0.0153
-1	0.0640	0.0406	-0.0235
0	0.0631	0.0315	-0.0316
1	0.0400	0.0223	-0.0178
2	0.0677	0.0532	-0.0145
3	0.0796	0.0538	-0.0258
4	0.0549	0.0508	-0.0041
5	0.0023	-0.0034	-0.0057
6	0.0167	0.0152	-0.0015
7	0.0155	0.0275	0.0120
8	-0.0116	-0.0240	-0.0125
9	-0.0792	0.0018	0.0810
10	-0.0684	0.0055	0.0740
11	-0.0482	-0.0357	0.0125
12	-0.1086	-0.0678	0.0408

Lag	Manual	Texttract	Difference
-12	0.2990	0.2919	-0.0072
-11	0.3700	0.3638	-0.0062
-10	0.4427	0.4373	-0.0054
-9	0.5025	0.4994	-0.0032
-8	0.5564	0.5535	-0.0029
-7	0.6161	0.6112	-0.0049
-6	0.6742	0.6710	-0.0032
-5	0.7127	0.7082	-0.0045
-4	0.7570	0.7543	-0.0028
-3	0.7924	0.7857	-0.0067
-2	0.8258	0.8146	-0.0113
-1	0.8714	0.8703	-0.0011
0	0.8990	0.8974	-0.0016
1	0.8505	0.8448	-0.0057
2	0.7992	0.7878	-0.0114
3	0.7582	0.7448	-0.0134
4	0.7265	0.7115	-0.0149
5	0.6898	0.6734	-0.0164
6	0.6325	0.6177	-0.0148
7	0.5645	0.5511	-0.0134
8	0.4762	0.4643	-0.0119
9	0.3993	0.3896	-0.0097
10	0.3410	0.3265	-0.0145
11	0.2643	0.2457	-0.0186
12	0.1882	0.1677	-0.0205

Author Contributions

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Ethics Statement

This study used aggregate historical public-health records published by the Dominion Bureau of Statistics, comprising weekly counts of notifiable disease cases by province and territory. The data contain no individual-level or identifiable information, and no human participants were involved. Institutional review board approval was therefore not required.

Data Availability

The Canadian Disease Incidence Dataset (CANDID) used in this study is publicly available through McMaster University’s Theoretical Biology group [11]. All Python scripts used for data extraction and post-processing are available as supporting information accompanying this manuscript. Data underlying all figures and tables reported in this manuscript are available in the supporting information files.

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⁹⁹⁴ **Competing Interests**

⁹⁹⁵ The authors have declared that no competing interests exist.